

BioSpecInfo

Dossier completo

Samuele Pio Provenzano

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Raccolta integrale dei documenti del progetto, nell'ordine di numerazione. Ogni documento è disponibile anche come PDF separato.

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BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	RDKit cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as PWA	Spectra Centre : IR, ¹ H/ ¹³ C-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and 3D (three.js, 3Dmol.js) rendering	Organic Chemistry : 25 interactive modules
Service Worker offline-first, versioned	Retrosynthesis : 46 strategies + animated 2D/3D mechanisms

Technical architecture	Content & capabilities
Zero copleft dependencies (MIT/BSD-3 only)	3D Drug Atlas: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **¹H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ± 0.5 ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm⁻¹** (90% within ± 30 cm⁻¹)
- Interactive performance **30-60 fps** from iPhone/iPad to Raspberry Pi 4

3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, Brenk/PAINS structural alerts, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

4. Technology stack

HTML5 · ES6/ESNext · WebAssembly (RDKit MinimalLib, sql.js/SQLite) · WebGL (three.js, 3Dmol.js) · SVG / Canvas 2D · Service Worker / PWA · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.
1. The reported validation data (MAE 0.31 ppm for ^1H -NMR vs SDBS; MAE 11 cm^{-1} for the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.
1. The architecture is **local-first**: no user personal data is transmitted to author-controlled servers (alignment with GDPR principles).

This is an **author's declaration**, not a certification issued by a third party.

6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
 2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
 3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
 4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.
- _Software license: proprietary — "All rights reserved" (see [LICENSE](#))._

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	<code>samupropiol-ship-it/BioSpecInfo-v11</code>

1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

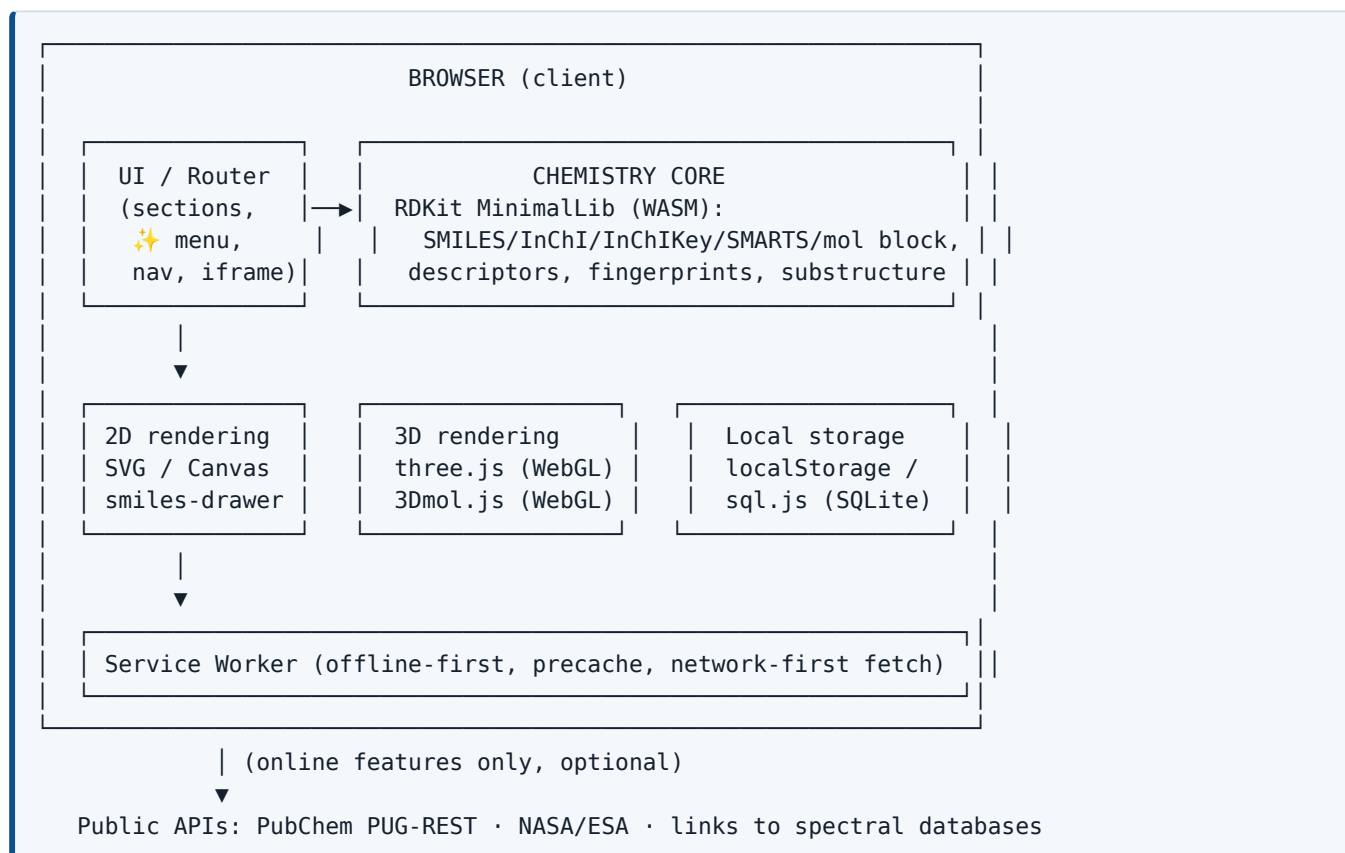
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

2. File structure

File	Role
<code>index.html</code>	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
<code>chimorga.html</code>	Organic Chemistry course (25 modules).
<code>astro.html</code>	3D Astrochemistry (three.js).

File	Role
pro.html, rdkit_lab.html, accademia.html, simulazioni.html, guidaret.html, file_manager.html, Biochimica_Guida_Definitiva.html, sr_completo.html, sr_essenziale.html, download.html, changelog_tesi.html	Thematic sub-apps loaded in iframes with <code>?v=timestamp</code> cache-busting.
manifest.json	PWA configuration (icons, theme, standalone).
sw.js	Service Worker: <i>network-first</i> strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
RDKit_minimal.js / RDKit_minimal.wasm	Cheminformatics engine (RDKit MinimalLib).
3Dmol-min.js	3D viewer for molecules/macromolecules.
three.min.js, three_bloom.js, gltf_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
smiles-drawer.min.js	2D depiction of SMILES.
lib/sql-wasm.js / lib/sql-wasm.wasm	SQLite in WebAssembly (SQL playground).
models/, textures/	3D assets (organs, écorché, planets) and textures.

3. Component architecture



3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframes`.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS, MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP (Wildman-Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp³, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg` , `get_svg_with_highlights`) and 2D coordinate generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D (molecule editor, CIP stereochemistry viewer), smiles-drawer.
- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito / blocked-storage scenarios).
- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.

- **Predicted spectra** as sub-panels: IR, ^1H -NMR, ^{13}C -NMR, MS with ****isotope pattern (IUPAC abundances) and adduct masses**** ($[\text{M}+\text{H}]^+$, $[\text{M}+\text{Na}]^+$...), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET**: Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry ([chimorga.html](#))

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (sql.js), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + [SharedArrayBuffer / Atomics](#) .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR; hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
- Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.

6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real

3D models.

- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	<code>node --check</code> on the critical JavaScript blocks on every change.	✓ In use
End-to-end (E2E) testing	Automation with Playwright on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: 0 non-network errors).	✓ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✓ In use
Manual cross-device testing	Testing on real devices (see §4).	✓ In use
Automated unit-test suite (Vitest/Jest)	—	 Roadmap
Coverage ≥ 90%	—	 Roadmap

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	SDBS (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	0.31 ppm
Signals within ± 0.5 ppm	89%

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	NIST Chemistry WebBook
Mean Absolute Error (MAE)	11 cm⁻¹
Bands within ± 30 cm ⁻¹	90%

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol* $\rightarrow$ *C₂₇H₄₆O*, 4 rings, 8 stereocentres.

Benchmark data source: measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30-60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30-60 fps

Size of the main `index.html` file: $\approx$ **4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

- Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.
 - Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure-spectrum relationships, not to replace certified experimental data.
 - Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.
 - Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.
-

6. Release acceptance criteria

- ✓ `node --check` passes without errors on the modified modules.
 - ✓ Playwright E2E: 0 non-network JS errors.
 - ✓ No visible regression in the touched sections.
 - ✓ Service Worker cache version bumped.
 - ✓ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
 - ✓ `verifica_guida`: whatever the documentation promises exists in the code.
-

7. Verification battery — state at version `bsi-v181`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	<code>verifica-farmaci</code> , <code>test_spettri</code> , <code>test_assi</code> , <code>test_assi_canvas</code> , <code>test_costanti</code> , <code>audit_dati</code>	structure vs molecular weight, functional-group recognition, axis conventions, physical constants

Family	Benches	Subject
AI agent	test_ko, test_404, test_503, test_firma, test_attesa, test_attesalunga, test_tetto, test_nucleo, browser_ko, browser_prova	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	audit_stabilita, audit_promesse, audit_quota, test_filemanager, test_doppioinvio, test_sw, test_visore3d	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	browser_reset, browser_proxy, browser_proxyui, browser_rdkit, browser_lab, browser_frontiera, test_aggiorna, test_guidaproxy	panels, proxy, RDKit laboratory, updates
Consistency	verifica_guida	documentation ↔ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	89 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, +0 on the four that follow
Survival of exhausted storage	localStorage.setItem forced to throw	10 pages out of 10 stay operational
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in 3,507 ms (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to 1

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.
- Probing `localStorage` exhaustion with a one-byte write proves nothing: after saturation, one byte always finds room.
- The ticks of an axis drawn on `<canvas>` are not readable as text: they must be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.

Security, Privacy & Compliance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Model	Client-side / Local-First PWA

1. Privacy & data protection (GDPR)

1.1 Local-First by design

- **No account, no login, no advertising tracking.**

• All user data (drawn molecules, preferences, quiz history, study progress) is stored **exclusively locally** on the device (`localStorage` / IndexedDB / `sql.js`), with no transmission to author-controlled servers.

- **There is no application backend** collecting or processing personal data.

1.2 Data handling and GDPR alignment

- **Data minimisation:** the app requires no personal data to function.

• **No transfer to the author's third parties:** data never leaves the device for infrastructure controlled by the author.

• **Optional, transparent external calls:** only upon a user action (searching a molecule, opening a 3D model) does the app query public third-party APIs (PubChem, NASA/ESA, spectral databases). Such requests contain **only the chemical identifier** (e.g. SMILES/name) and **no personal data**. Use of these services is subject to their respective privacy policies.

• **Local right to erasure:** the user can delete data by clearing the site storage in the browser.

Note. This alignment describes the local-first design properties that favour GDPR compliance. It does not constitute a legal certification; for regulated contexts a dedicated DPIA assessment is recommended.

2. Application security (OWASP)

2.1 Attack surface

Being a **static application with no backend**, entire classes of server-side vulnerabilities are absent by construction (SQL injection on a server DB, server-side RCE, server authentication/session, IDOR on proprietary APIs).

2.2 Input sanitisation

- Parsing of **SMILES / SMARTS** strings and molecular files is delegated to

RDKit MinimalLib (WASM), which validates the structures and rejects malformed input by returning invalid molecules rather than executing them.

- Strings coming from user input or external APIs, when inserted into the DOM, are treated as untrusted. Their textual value (names, notations) is shown as text content/ `textContent` or with escaping of the `& < >` characters.

2.3 Runtime robustness

- **localStorage guard**: every storage access is protected by `try/catch` with an in-memory fallback, so that incognito mode or blocked storage cannot halt execution (a historical regression that has been fixed).
- **Controlled degradation**: panels never silently stay empty; errors show a message and, where applicable, a "Retry" button.
- **Sub-app isolation** in `iframe`s.

2.4 Content Security Policy (deployment recommendation)

Since the app can be published on static hosting, it is recommended to serve it with security headers at the hosting/CDN level:

- Restrictive `Content-Security-Policy` (self script/style + WASM),
`frame-ancestors`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`.
- `Cross-Origin-Opener-Policy` / `Cross-Origin-Embedder-Policy` if `SharedArrayBuffer` and parallel Web Workers are enabled in the future (HPC roadmap).

3. Integrity and reproducibility (GAMP 5 alignment)

For use in pharmaceutical/regulated contexts, the architecture is **compatible** with GAMP 5 principles:

- **Determinism**: the cheminformatics computations (RDKit) are deterministic and reproducible for the same input.

- **Version traceability:** code versioning on Git, versioned Service Worker cache, changelog.

- **Method transparency:** the heuristic predictors are documented with their sources and limitations (see the *Validation Report*).

Again: this is **alignment with the principles**, not a formal qualification (IQ/OQ/PQ), which would require a dedicated validation process at the using organisation.

4. Summary

Aspect	Status
Personal data transmitted to author's servers	None
Backend / server database	Absent
Molecular input validation	RDKit WASM
DOM output escaping	Yes (untrusted content)
Storage guard / error degradation	Yes
CSP / security headers	Recommended at hosting level
Offline operation	Yes (Service Worker)

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	RDKit_minimal.js , RDKit_minimal.wasm	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	BSD-3-Clause
3Dmol.js	3Dmol-min.js	3D viewer for molecules/macromolecules (WebGL)	BSD-3-Clause
three.js	three.min.js , three_bloom.js , glTF_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	MIT
SmilesDrawer	smiles-drawer.min.js	2D depiction of SMILES strings	MIT
sql.js (SQLite compiled to WASM)	lib/sql-wasm.js , lib/sql-wasm.wasm	Client-side SQL playground	MIT

The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms

Service	Use
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

3. Intellectual property of the proprietary code

- BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine, the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.
- No copyleft-licensed code** is embedded in the proprietary source: the author retains full licensing freedom over their own code.

3.1 Project license

The repository includes a **proprietary — "All rights reserved"** [LICENSE](#) file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	None
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project LICENSE file	 Present — proprietary "All rights reserved"

AI Agent Architecture "Spectra" — BioSpecInfo

Field	Value
Project	BioSpecInfo — Spectra component (agentic AI copilot)
Author	Samuele Pio Provenzano
Thesis supervisor	Prof. Savino Longo — University of Bari Aldo Moro
Component	<code>bsi-ai-hub.js</code> — 6,254 lines, zero runtime dependencies
Type	Multi-provider conversational agent with client-side tool execution
Repository	<code>samupropio1-ship-it/BioSpecInfo-v11</code>
Documented version	Service Worker <code>bsi-v181</code>

1. Executive summary

Spectra is an **agent**, not a thin wrapper around a language model. The difference is measurable: the model does not answer quantitative questions from memory — it **invokes deterministic tools** that compute the result, and the control loop lets it **chain up to 10 calls**, verifying each step before answering.

The architecture addresses three problems that separate a usable agent from a demo:

1. **Grounding** — a fabricated chemical value can be dangerous. 35 tools cover computation, public databases and the application's own datasets; the system prompt explicitly forbids quoting numeric values from memory.
1. **Transparency** — the model's reasoning is streamed live and stored alongside the answer, together with the list of tools actually invoked. The agent's work is auditable after the fact.
1. **Provider portability** — a single tool registry is translated into the three function-calling formats in use today (OpenAI-compatible, Anthropic, Gemini), so the agent behaves identically across ten model configurations, four of them free.

2. Architecture

2.1 Overview



There is no backend. API keys stay in the browser's `localStorage` and travel only to the selected provider: no data passes through BioSpecInfo servers, which do not exist.

2.2 Agentic loop

Each round: send the conversation history (last 40 turns) plus the system prompt and the schema of all 35 tools → receive the streamed response → if it contains tool calls, execute **all** of them → rebuild the assistant turn in the provider's native format → repeat.

Three exit conditions: no tool calls (final answer), `MAX_ROUNDS = 10` exhausted, or user interruption.

2.3 Provider-neutral tool registry

Tools are declared once in JSON Schema and translated at runtime:

Family	Required shape
Anthropic	<code>{name, description, input_schema}</code>
Gemini	<code>[{functionDeclarations: [...]}]</code>
OpenAI-compatible	<code>{type:"function", function:{...}}</code>

The same applies to messages: turns containing tool calls are serialised into each provider's native form and kept in a `_native` field, so a conversation stays coherent even when switching models.

2.4 No hard-coded model names

A model name written into the code is a time bomb: it works until the provider retires that model, and then the application stops with a 404 the user cannot fix. It has happened twice in this project:

Provider	Error	When
Google	<code>models/gemini-1.5-flash</code> is not found for API version <code>vlbeta</code>	removed from <code>vlbeta</code>
Groq	The model <code>llama-3.3-70b-versatile</code> does not exist or you do not have access to it	deprecated 17/06/2026

On some services the problem is structural: free model ids change constantly, by design. (That is why OpenRouter was later dropped from the list, see 2.4.)

The fix is not to write the new name — that would be the same bomb with a longer fuse — but to **remove the name** and ask the API, which knows what exists *for that key*. This also resolves the ambiguity in Groq's message: "does not exist" **or** "you do not have access" are different cases, and a per-key listing tells them apart.

Every configuration with `modelIdCandidati` starts at `model: null` and is resolved at runtime. The two families use different strategies because their names are different:

Gemini — version scoring. Names follow a regular scheme (`gemini-<major>.<minor>-<family>`), so they can be ordered: newest version first, `flash` family preferred, with penalties for experimental, `preview` and `-lite` variants; models without `streamGenerateContent` are discarded, as are non-conversational ones (embedding, image, native audio, Live API). When Gemini 3.0 ships it will be picked automatically.

OpenAI family — candidates in preference order. Here names are not comparable to each other (`openai/gpt-oss-120b` versus `qwen/qwen3.6-27b`) and automatic scoring would choose badly. `GET /models` is queried and the **first candidate that still exists** wins: the order of the list *is* the preference, and the listing serves to skip the ones that vanished. Only if no candidate survives does a generic score run over the available models — discarding transcription, speech, embedding and guard models, and preferring the larger model all else being equal.

Three fallback levels in both cases: model listing → candidate probing → first candidate, letting the real error from the generation call speak instead of inventing one.

The choice is cached for seven days per provider, keyed to a 32-bit FNV-1a fingerprint of the API key (different keys see different catalogues; the key itself is never duplicated). If the model is retired *while* cached, the `404` — or a `400` whose text mentions the model, as some providers return — invalidates the cache, re-resolves and retries **once**: a flag prevents an infinite loop when the new model fails too.

Anthropic is the deliberate exception: its models are paid, explicitly chosen by the user, and deprecated with long notice.

The three free services, chosen one by one

Free quality is not all alike, and the difference matters: an 8-billion-parameter model does not hold up on a multi-step physical-chemistry problem. The list has been **shortened**, not extended — two services were removed because they made answers worse, not better.

Service	Models	Its role
Groq	GPT-OSS 120B, Qwen3	The workhorse: 131K context, ~30 requests/minute
Google Gemini	Gemini Flash	Up to 1M context: whole documents and PDFs
Z.AI GLM	GLM-4.7-Flash	Free with no expiry : the reserve still there when credits are gone

Removed. *GitHub Models*: it was here, and was removed because GitHub **retired it entirely on 30 July 2026** — playground, catalogue and inference API switched off for everyone. It had been added two days earlier on the strength of pages describing the still-live service: the lesson is that verifying *a thing exists* is not verifying *it is still alive*, and the search to run is "<name> retired / shutdown / deprecated <year>". The failure did not stop at the dropdown entry: the rank assigned to it also made it Core Mode's automatic choice.

Mistral: mid-tier quality and a free plan requiring consent to data training — not worth paying for unpublished thesis material when five alternatives carry no such clause. *OpenRouter*: small free models with ids that change constantly, unsuited to an agent chaining ten tool calls. *Cerebras*: since August 2026 there is no card-free plan, and free context is capped at 8K — Spectra sends 32 tool definitions on top of history, which does not fit.

None of the five matches a paid frontier model on the hardest problems; GitHub Models come closest, at the cost of low request ceilings. That is why the choice stays the user's rather than imposing one service.

The paid frontier

Free tiers are enough for studying; for a hard problem they are not. The selection criterion was **GPQA Diamond** — graduate-level physics, chemistry and biology questions — because it is the only benchmark measuring what this app actually asks.

Service	Why it is here
GPT-5.6	94.6% on GPQA Diamond: the highest score available today. ~\$4/M input
Gemini 3 Pro	Over 1M context, and the same key as free Gemini. ~\$2/M
DeepSeek V4	High-tier reasoning at ~\$0.66/M: the best quality/price ratio
Grok 4.6	First on agentic tool calling and lowest hallucination rate — exactly the profile an agent chaining tools needs. ~\$2/M
Claude (Fable 5.1, Opus 5, Sonnet 5)	One key for all three; the only ones with the Python sandbox in Core Mode

Two configurations can use **the same key**: the four Claude models are one Anthropic account, and Gemini Flash and Gemini 3 Pro one AI Studio key. `chiaveCondivisaCon` points them at the same entry, so the user pastes it once. Clearing it clears it for all twins — leaving one behind would make the key reappear on the other.

The two Gemini entries forced the scorer to be generalised: they share family, endpoint and key but want **opposite** models. The preference is now declared in configuration (`/flash/` versus `/pro/`) and outweighs the rest of the score. Without it the free configuration would have picked `gemini-3-pro` — a newer generation, therefore a higher score, but **paid**: a model the free key cannot use.

The same pass surfaced a second issue: the minor version in the name is **optional**. From generation 3 Google dropped it (`gemini-3-pro` , not `gemini-3.0-pro`), and the regex requiring the dot scored the *newest* models as generic aliases — so they lost to older ones, and the automatic upgrade described in 2.4 would never have triggered.

The provider audit, and why it must be repeated

The system repairs itself on **model names** — it queries the catalogue and picks among what exists. It does not repair itself against a **dead service**: that must be checked by hand, and GitHub Models' retirement showed what not doing so costs.

September 2026 check, provider by provider:

Provider	Result
Groq	Active. <code>llama-3.3-70b-versatile</code> is no longer served as of August 2026 and <code>llama-3.1-8b-instant</code> is deprecated: removed from candidates, leaving Groq's own replacements
Google Gemini	Active
NVIDIA NIM	Removed 09/2026: alive, but without CORS headers it cannot be called from a web page
Z.AI	Active, GLM-4.7-Flash and 4.5-Flash still free
OpenAI, DeepSeek, Anthropic	Active
xAI	Active, but candidates were two generations behind: the top is <code>grok-4.6</code> since 12/08/2026

Two traps recorded so they are not repeated. First: never put a paid model among a free configuration's candidates — true for `glm-5.x` on Z.AI as it was for `gemini-3-pro` , or the configuration picks something the free key cannot use. Second: a dead provider with a high rank does not merely break a dropdown entry, it **becomes Core Mode's automatic choice**.

When a provider does not answer

A `fetch` that fails **before** receiving a response means one of three things: no network, the service does not accept direct browser calls (CORS), or the service no longer exists. The browser, for security reasons, does not let you tell which — the error is deliberately generic.

This used to stop the turn. But if the three causes cannot be told apart, the right response is the same for all three: **move to another provider** the user holds a key for, exactly as for exhausted quota. It is the behaviour that would have masked GitHub Models' retirement entirely, instead of leaving every request stuck on a dead service.

Errors that would repeat identically everywhere stay out of the fallback — a wrong key, a malformed request: trying them on every provider and then reporting a random last error would hide the real cause.

Removing a service from the list is a breaking operation, and deserves a note: whoever had it selected has that name saved in `localStorage`, and without validation would hit `PROVIDERS[undefined]` — Spectra would no longer open. `getSavedProvider()` always validates against the registry and falls back to an existing service.

2.5 The optional proxy — Spectra without a key

The API key lives in the user's browser: with no backend there is no alternative, and it is the limitation stated openly at the end of this document. Every visitor has to obtain their own free key and paste it in.

`proxy/spectra-proxy.js` is the way out, and it sits **outside** the page: a Cloudflare Worker (free plan) holding the keys as server-side secrets. When it is configured the browser carries no key at all, and anyone opening BioSpecInfo uses Spectra without entering anything.

It is not a plain forwarder:

- **The path passes through untouched.** The route is `/<provider>/<path>` and the rest is relayed as received: the proxy knows nothing about API shapes and keeps working when Spectra changes model or parameters.
- **Several keys per provider, with automatic takeover.** Each secret is a comma-separated list; on `429` (quota) or `401/403` (revoked key) it moves to the next one within the same request. Not on `400` — that is a bad request and would fail identically.
- **Streaming is never buffered:** the response body is passed straight through, otherwise answers would appear all at once at the end.
- **It is not an open relay.** Credentials arriving from the client (`Authorization`, `x-api-key`, `?key=`) are discarded and replaced: nobody can use the proxy to push a key of their own.
- **Three brakes** against abuse: an allowed-origins list, a per-IP limit and a daily cap.

On the Spectra side the graft is in one place. `GET /stato` reports which providers the proxy actually covers, so a model whose secret is missing is never offered as "no key needed"; `buildRequest` routes to the proxy **and omits authentication entirely**; `chiaveDaUsare()` returns a placeholder so the "missing key" guards do not block sending — a placeholder that never leaves the browser.

Both paths coexist: without `PROXY_URL` everything works as before, and for providers the proxy does not cover Spectra keeps using the local key.

2.6 Starting over

Everything Spectra remembers lives in `localStorage`: chats, history, API keys, persistent memory, review cards. A "clear history" button that removed only the conversations would not start over — reopening the app would restore the same state.

The delicate part is that keys exist in **two formats**: the current map (`bsi_api_keys`, one key per provider) and the single-key format of earlier versions (`bsi_api_key`). `getKeysMap()` migrates the latter into the former, so clearing only the map **brings the old key back** on the next access. Both must go — along with the resolved-model caches, which mean nothing without a key.

That is why the list of everything the application writes lives in one place, `DATI_CANCELLABILI`, split into five groups with a label and a description. The confirmation panel is built from it: it shows how many entries actually exist per group, disables the empty ones and requires a second explicit click. Only chats and keys are pre-selected; memory, review scheduling and the rest of the app's progress must be chosen deliberately, because they erase work unrelated to starting the conversation over.

Deliberately excluded: the PRO licence, the trial period and the device identity — data a user does not expect to lose by emptying a chat, and a licence must never be deleted by accident. The panel says so.

After deletion the in-process state is reset too — the list of providers the proxy covers, the resolved model per provider — otherwise it would stay valid until the page reloads, and Spectra would keep using a model chosen with a key that no longer exists.

2.7 Making free keys carry real load

Free tiers impose two different limits, and each needs a different answer.

The per-request ceiling. GitHub Models accepts 8,000 input tokens. Spectra's fixed cost is ~8,150 — 2,009 for the system prompt plus **6,128 for tool definitions alone** — so that service would not even start: it would fail before the user typed a word. The measurement came before the fix, and changed what the fix had to be.

The answer is not to trim history — it is to trim **the fixed cost**. Tools are paid on every turn; the conversation is the content. Of the two, the fixed cost must shrink first. `adattaAlBudget()` selects the tools **relevant** to the question instead of all 32, then shortens history with what remains.

Relevance is scored against a keyword map kept **outside** the tool registry: the words people use to *ask* for something are not the words a tool is *documented* with. The description of `astrofisica` talks about Wien and Stefan-Boltzmann; the user writes "what temperature is this nebula". Matching the question against descriptions alone picked `farmacocinetica` for an astrochemistry question — measured, not assumed. Matching is by stem (first five characters) because Italian inflects: "nebulosa" and "nebulose" must count as the same word.

Rate limits. A 429 on a free tier is almost always the *per-minute* limit, not an exhausted quota: waiting a few seconds clears it. The provider says how long in `Retry-After`, or in the error text ("*Please try again in 7.5s*"); when it says nothing, exponential backoff with a little jitter, so every open tab does not retry at the same instant. If the requested wait is absurd, it does not wait at all: better to say "quota exhausted" than to leave the user watching an hourglass for an hour.

Provider fallback. When a quota really is exhausted, the turn is redone on another service the user holds a key for. This is why several free keys together carry load that none of them carries alone.

Two non-negotiable constraints:

- It restarts from the **original** messages, not from half-finished history.

Turns containing tool calls are stored in the previous provider's native format and cannot be handed to another. The turn's work is lost; a correct answer is gained.

- Fallback uses **free services only**. Silently moving to a paid one would spend the user's money without their say-so; if the chosen service was paid, it still gets the first attempt.

And it moves on **only** for exhausted quota. A 401 or a malformed request would repeat identically everywhere: trying all providers and then reporting a random last error would hide the real cause.

2.8 Core Mode, and saying what is happening

The mode. One switch that changes four things at once, because none of them alone would move the needle: it switches to the most capable configuration among those with a key, raises agentic rounds from 10 to 16 (a long tool chain ran out halfway), requests maximum reasoning depth from models that expose it, and adds a system-prompt instruction to work in verified steps rather than answer fast.

The model switch is **announced in the chat**, and if the new one is paid it says so: switching silently to a metered service would spend the user's money without telling them. For the same reason it is a switch and not the default — it costs more quota, and on a paid service it costs money.

Saying what is happening. Several seconds can pass between sending and the first word of the answer: the model reasons, calls tools, waits on the network. Before, the user saw a mute pause and concluded it had frozen.

The header now carries an atomic core whose orbitals change speed and colour with state — slow and cyan at rest, fast while reasoning, amber while a tool runs — alongside a line saying the same thing in words, for anyone not watching the animation. The states hook into callbacks the agentic loop already emitted: no mechanism was added, only made visible.

Three layout defects fixed in the same pass, all found by measuring rather than looking:

- The input field shared its row with four buttons and stayed 156 px wide: the

placeholder wrapped and was **clipped**. The text now has its own row with the controls below it. The placeholder length was chosen by testing at 360 px width, not by eye.

- The title "Spectra — il copilota AI di BioSpecInfo" did not fit and truncated mid-word. Only the name remains; the rest became the status line, which now has a purpose.
- The core's orbitals, when rotating, escaped their box and overlapped the name.

2.9 Using what the provider actually offers

Three things the Anthropic API provides that were not being used. Checked against the official reference rather than recalled — two of the three carry a constraint that would not have been guessed.

Prompt caching. The fixed cost of every request is about 8,150 tokens: 2,000 of system prompt plus 6,128 of tool definitions, identical every turn. Marked for caching they are paid once and then read back at a fraction of the price.

The constraint is that caching works by *prefix*: one differing byte at the start invalidates all of it. But Spectra's prompt also carries user memory and question context, which change every message. Concatenated into one string the cache **would never have hit**. The prompt now travels in two blocks — stable and marked, volatile and not — and the marker also goes on the last tool, because composition order is `tools` → `system` → `messages`.

The Python sandbox, and why it excludes web search. `code_execution` gives the model an environment with sympy, numpy, scipy and matplotlib: numerical integration, symbolic algebra, matrix diagonalisation — things the 32 solvers do not cover because they cannot all be anticipated.

But the new-generation web search **already carries an execution environment** (it filters results by writing code), and declaring `code_execution` as well would create a second one, confusing the model. They are alternatives, not complements. The choice follows Core Mode: normally web search and page fetching, in Core Mode the sandbox — because when maximum power is requested the problem is computational, not documentary. A test verifies the two never appear together.

Page fetching. `web_fetch` joins search: if the user pastes an article link, Spectra reads it instead of merely searching for its title.

Sandbox results arrive as `bash_code_execution_tool_result` and `text_editor_code_execution_tool_result` — not a generic `code_execution_tool_result` — and must be kept in the assistant turn like the web-search blocks, or a server-paused turn cannot be resumed.

3. The 35 tools

The list is the one **measured** in `window.BSI_AI_T00LS` on the running application, not transcribed from memory: this document declared 32 when there were already 35, and three — `analizza_molecola`, `disegna_molecola`, `mostra_spettri` — did not appear at all.

Area	Tools
General computation	calcola, risolvi_equazione, analisi_dati
General chemistry	bilancia_equazione, stechiometria, massa_molecolare, converti_unita, costante_fisica
Physical chemistry	termodinamica, equilibrio_acido_base, cinetica, gas_e_soluzioni, elettrochimica
Structure & spectra	spettroscopia, quantistica_e_spettroscopia, cristallografia, analizza_molecola, disegna_molecola, mostra_spettri
Life sciences	biochimica, farmacocinetica, valuta_druglikeness
Physics	astrofisica, nucleare, statistica_inferenziale
External databases	cerca_pubchem (NIH), cerca_letteratura (PubMed)
Internal data	cerca_nel_database (9 datasets), cerca_molecola
App control	naviga_sezione (89 sections), apri_strumento (12 labs), stato_app
Memory	ricorda, ricordi
Animations	apri_animazione (6 reaction mechanisms)

Web search is not among these thirty-five, and the distinction matters. The thirty-five are executed **by the application**, in the user's browser. On providers that offer them, the agent adds two tools to the schema that are executed **by the provider** — `web_search` and `web_fetch`, with a per-turn usage cap. Counting them with the others would make them look always available, whereas they depend on the provider chosen.

3.1 Internal datasets exposed

297 synthesis reactions · 118 elements · 67 amino acids · 233 drugs · 63 pathologies · 46 retrosynthetic strategies · 36 drug interactions · 29 metabolic pathways · 29 redox potentials.

Drugs and strategies are the two figures that `tools/verifica-affermazioni.js` measures against the running application on every battery run: here they had been left at 143 and 39 while the app counted 178 and 46.

These are the datasets curated by the author for the application: the agent treats them as a primary source and is instructed to **report discrepancies** between the database and its own memory rather than silently smoothing them over.

4. Notable engineering decisions

This section documents the non-obvious choices and the reasoning behind them.

4.1 Expression engine without `eval()`

Problem. Serving arbitrary physical-chemistry computations requires an expression evaluator. `eval()` would solve it in a few lines.

Why it was rejected. The agent reads untrusted external content (PubChem results, PubMed abstracts, web pages). An injection in that content could induce the model to emit a hostile expression; with `eval()` that string would execute in the page context, **where the user's API key lives**.

Solution. Recursive-descent parser with a *closed* set of 27 functions and constants. It has no access to global scope: it can only do arithmetic. Validated against 10 escape attempts (`window.localStorage`, `constructor`, `this`, `fetch(...)`, calls to non-whitelisted functions) — all rejected, including when executed in a real browser.

4.2 Preserving reasoning blocks

On models with adaptive reasoning, `thinking` blocks are part of the assistant turn and must be **echoed back unchanged, cryptographic signature included**, on the next tool-use round. Dropping them causes the request to be rejected.

The implementation accumulates `thinking_delta` and `signature_delta` from the stream and reinserts them **first** in the reconstructed turn. Verified in integration tests: signature preserved, ordering correct.

4.3 Resuming paused turns (`pause_turn`)

Web search is a server-side tool. When its internal loop exhausts its iterations, the response ends with `stop_reason: "pause_turn"`.

The turn must be resumed by **sending the assistant turn back as-is, without adding any user message**: the server recognises the trailing `server_tool_use` block and continues from there. Appending a "continue" message would break the mechanism.

4.4 Per-model configuration, not global

The four Claude tiers do not accept the same parameters:

Model	effort	Web search	Notes
Fable 5.1	<code>xhigh</code>	yes (8)	refusal fallbacks; reasoning always on
Opus 5	<code>high</code>	yes (6)	default
Sonnet 5	<code>high</code>	yes (5)	
Haiku 4.5	unsupported	unsupported	sending them returns HTTP 400

On all recent models `temperature`, `top_p` and `budget_tokens` were removed from the API and their presence causes an error: none of the four sends them. Every parameter is therefore conditioned on the provider rather than set globally.

4.5 Multimodal input and material reading

The agent accepts images, PDFs and text files, several at once, treated as consecutive pages of a single document. Three constraints shaped the implementation:

- **Images are downscaled to 2000 px** and re-encoded as JPEG before sending.

A phone photo drops from roughly 1 MB to 440 KB: without this step a single image would approach the request ceiling and multiply token cost. 2000 px is the minimum for reading small handwriting — at 1600 px fine script became illegible.

- **The PDF page ceiling depends on the model**, it is not a constant: 600

pages for models with a 1M-token window, 100 for 200K ones. Page counting is done by reading `/Type /Page` objects from the file bytes, without libraries.

- **History stores a 160 px thumbnail**, not the image that was sent. Storing

the large one (440 KB each) filled `localStorage` within a dozen photos, and once the quota is exceeded every subsequent write fails silently.

Six quick actions turn the material into a study artefact (summary, concept map, outline, flashcards, exam questions, transcription) and two translate it. For transcriptions the agent is instructed to write `[illegible]` rather than guess: an invented chemical term is worse than a declared gap.

4.6 In-house graph renderer, no CDN

Concept maps are produced by the model as Mermaid diagrams and drawn by a renderer written into the application (~190 lines): longest-path layering from the roots with cycle breaking, barycentre reordering over three passes to reduce edge crossings, SVG output with Bézier curves.

Not loading Mermaid from a CDN follows from the app's offline-first nature: a map must be drawable without a network. Unsupported syntaxes are recognised and left as a readable code block instead of being drawn badly.

4.7 Voice command with wake word

User-activated continuous listening: the phrase "Hey Spectra" followed by a command sends it automatically. Two non-obvious details:

- the browser's speech recognition **stops on its own** after a few seconds of

silence and must be restarted, otherwise listening dies at the first pause;

- **all transcription alternatives** are examined, not just the first, and the

most common phonetic variants are accepted: Italian recognition renders "Spectra" in many different ways.

Denying microphone permission stops the loop instead of retrying forever. The word "spettroscopia", extremely frequent in this domain, was verified as non-triggering.

4.8 Guard against non-finite results

A systematic audit revealed that, given legitimate degenerate inputs (zero wavelength, zero volume, zero half-life, a transition between identical levels), eight solvers returned `ok: true` with `Infinity` or `NaN` among their fields.

This is the most insidious failure in this context: an exception is noticed, a formally valid but meaningless value is reported to the user as correct. The fix is a single check at the point where all results pass through: if a numeric field is not finite, the result becomes an explicit error naming the fields and stating the probable cause. It also covers tools added in the future.

4.9 Graceful degradation

- **Models without reliable function calling** (some free tiers): on the first tool-related error the request is retried once in text-only mode instead of failing.
- **Idle timeout (45 s)**, reset on every byte received: a long answer is never truncated, but a dead connection is reported immediately with an explicit message.
- **localStorage quota**: history is capped (30 conversations, 100 messages) with progressive pruning. Without this cap, exceeding the quota made every subsequent write fail *silently*, **including saving the API key**.
- **Classifier refusals** (`stop_reason: "refusal"`, HTTP 200): detected and surfaced, instead of appearing as an empty answer.

5. Verification

Verification was carried out at three levels, with recorded outcomes.

5.1 Numeric correctness — against reference values

Area	Case	Expected	Obtained
Balancing	<code>KMnO4 + HCl → ...</code>	2:16:2:2:8:5	✓ exact
Molecular mass	<code>K4[Fe(CN)6]</code> (nested)	368.35	368.345
Molecular mass	<code>CuSO4·5H2O</code> (hydrate)	249.68	249.677
Equilibrium	acetic acid 0.1 M, K_a $1.8 \cdot 10^{-5}$	pH 2.874	2.875
Thermodynamics	ΔG (N_2O_4/NO_2) at 298 K	+4.79 kJ/mol	✓
Kinetics	E_a from $k(300\text{ K})$, $k(320\text{ K})$	55.3 kJ/mol	55.33
Real gases	van der Waals CO_2	−10 % deviation	−10.13 %
Spectroscopy	bromine M+2	97.3 %	✓
Biochemistry	glycylglycine	132.12 Da	✓
Pharmacokinetics	$t_{1/2}$ from V_d 50 L, Cl 5 L/h	6.93 h	✓

Area	Case	Expected	Obtained
Astrophysics	solar emission peak (Wien)	501.5 nm	501.52
Astrophysics	Earth escape velocity	11.19 km/s	11.186
Nuclear	⁵⁶ Fe binding energy	8.79 MeV/nucleon	8.790
Statistics	critical t, 95 %, df = 4	2.7764	✓
Crystallography	copper density (fcc)	8.96 g/cm ³	8.935

P-values are not tabulated: they are computed with the regularised incomplete beta function (Lentz continued fraction) and the incomplete gamma function.

5.2 Robustness — cases that must fail

Verified as **rejected**: unbalanceable equations, formulas with unbalanced parentheses or non-existent elements, species absent from the equation, invalid database types, and the 10 injection attempts against the expression engine.

5.3 Integration — real browser

Automated tests with headless Chromium across all 14 application pages: no JavaScript errors, no missing resources. All 89 sections of `index.html` and the 18 tabs of the Astrochemistry module were traversed one by one. Request bodies were inspected for the four Claude tiers, and a full cycle with web search, turn suspension and resumption was simulated.

6. Quantitative summary

Metric	Value
Component size	6,254 lines
Tools	32
Model configurations	10 (3 free)
Scientific areas covered	13
Records in exposed internal datasets	over 800
Max agentic loop rounds	10
History sent to the model	40 turns
Runtime dependencies	none

7. Stated limitations

Documented for technical honesty; these are deliberate choices, not omissions.

- **The API key lives in the browser** — unless the proxy is deployed. In

the default configuration the key never leaves the device, but it is accessible to anyone with access to that browser, and every user must obtain their own. The Worker in section 2.5 removes this limitation by moving keys to a server, at the cost of one component to maintain outside the page: a distribution choice, not a requirement.

- **Network tools depend on service availability.** PubChem and PubMed are queried directly from the browser; when unreachable, the tool returns an explicit error and the agent is instructed to say so rather than substituting an estimate.
- **Solvers apply simplified models** where the literature offers more refined ones (e.g. Woodward-Fieser for UV, or ATP yields using average P/O ratios). Each tool states the method used in its own result.
- **Verification is functional and numeric**, not formal: there is no proof of solver correctness, but a set of reference cases drawn from the teaching literature.

Companion document to the BioSpecInfo technical dossier. See also: [01-Software-Architecture-Document.md](#), [02-Verification-Validation-Report.md](#), [03-Security-Privacy-Compliance.md](#).

Scientific Accuracy and Data Provenance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	bsi-v181
Purpose	Document how the scientific data shown by the application are generated, by what method they are verified, and what the declared limits are.

Why this document exists. A chemistry teaching application can be impeccable from a software point of view and wrong from a scientific one, and the user has no way of noticing: a plausible molecular weight, a convincing-looking spectrum and a wrong structure all present themselves in the same way. This document describes the checks that make that error **visible and automatic**, and declares what stays beyond their reach.

1. Principle: every scientific datum must be open to contradiction

The criterion adopted throughout the application is that a scientific datum is not considered verified because it *looks* right, but because **a second, independent datum exists that would confirm or refute it**. Where that second datum does not exist, the limit is declared instead of being filled with an estimate presented as a measurement.

Three concrete applications of this principle:

Domain	First statement	Second, independent statement	Check
Drug database	structure (SMILES)	molecular weight from the literature	the weight recomputed from the structure must agree
IR / NMR spectra	functional groups recognised	molecular structure via RDKit	groups are sought on the graph, not on the text of the SMILES
Physical constants	tabulated value	symbol and unit of measure	the lookup refuses ambiguous matches instead of picking one

2. Pharmacological database — 233 entries

2.1 The verification method

Every entry declares **two** properties of the same molecule: the structure, in SMILES notation, and the molecular weight. They are independent statements — the first describes atomic connectivity, the second is a number taken from the literature — and therefore comparable:

weight computed by RDKit from the structure ↔ weight declared in the table

A discrepancy above **0.6 u** (a tolerance that covers rounding and small differences between atomic-mass tables, but not a real error, which is worth tens of units) indicates that at least one of the two statements is wrong.

The check is automated (`tools/verifica-farmaci.js`, runnable by anyone holding the repository) and requires no human judgement: it does not ask anyone to trust a source, it compares two numbers.

2.2 What it found

Applied to the 143 pre-existing entries, the check identified **89 errors**. The distribution is significant: the **molecular weights were correct** — taken from the literature — while the **structures were wrong**.

Drug	Structure declared
Ketamine	an unrelated pyrrolidinone
Morphine	non-matching structure (in two separate entries)
Lorazepam	missing the 3-position hydroxyl that defines its identity
Atorvastatin	missing the anilide group
Testosterone	missing the Δ^4 double bond of ring A

Plus **7 duplicated drugs** in different categories, with data that sometimes diverged between the two copies.

2.3 The discrepancy as a diagnostic instrument

The numerical value of the difference is not merely an alarm: it indicates **which** fragment is missing or superfluous.

Δ (u)	Interpretation	Real case
28.05	C ₂ H ₄	ketamine, fentanyl, morphine
16.00	one oxygen atom	tetracycline (one hydroxyl too many)
14.02	a CH ₂ group	doxycycline
2.02	a missing double bond	testosterone (Δ^4)
0.98	cytosine ↔ uracil	sofosbuvir

The last case deserves a note: a discrepancy of 0.98 u corresponds exactly to replacing an –NH group with an oxygen atom, that is, to the difference between cytosine and uracil. Sofosbuvir is a **uridine** analogue, and the proposed structure contained the wrong base. The number showed where to look.

2.4 Current state and declared limits

Drugs in the database	178 (it was 143)
With a verified structure	157 (it was 153)
Defects	0
Declared and accepted deviations	21 (it was 25)

Verification is **conformant**: no entry presents a structure that contradicts its own molecular weight. The 21 deviations are entries **without a structure**, each recorded with its own reason in `../evidence/deviazioni-note.json` and reported in the verification report.

Defect versus deviation. The check distinguishes two situations that must not be confused:

- a structure that **contradicts** its own weight is a defect, and fails

verification;

- an **absent** structure may be an acceptable deviation — but only if

explicitly recorded, with its reason. An entry without a structure and **not** recorded fails verification like any other defect.

Recording a deviation is therefore a deliberate decision, traced in git and visible in the report, not a way of silencing a check.

And it must also be **revoked** when it is no longer needed. From version `bsi-v181` the bench also fails in the opposite case: an entry listed in the registry that **does** have a verified structure is a permission left switched on for nothing, and tomorrow it would silently cover a wrong structure put in its place.

2.4-bis Four entries that left the registry

Of the six entries without a structure, four were closed by taking the structure from **ChEMBL** — a source independent of this project — and verifying it with the same structure ↔ molecular weight comparison that applies to all the others.

Entry	ChEMBL	Formula	Weight declared	Weight computed
Digoxin	CHEMBL1751	C ₄₁ H ₆₄ O ₁₄	780.94	780.95
Vincristine	CHEMBL90555	C ₄₆ H ₅₆ N ₄ O ₁₀	824.96	824.97
Tacrolimus topical (Protopic)	CHEMBL269732	C ₄₄ H ₆₉ NO ₁₂	804.02	804.03
Tacrolimus systemic (Prograf)	CHEMBL269732	C ₄₄ H ₆₉ NO ₁₂	804.02	804.03

The reason recorded for digoxin said "every structure tried departs by 14-30 u from the literature weight": the declared weight was right, it was the structures tried that were wrong. That the comparison is real and not complacent can be seen by adding a single carbon to the correct structure — the bench reports Δ 14.04 and fails.

The two that remain are not single molecules, and no SMILES notation represents them:

- **Ivermectin** is a mixture of homologues: at least 80 % B1a (C₄₈H₇₄O₁₄) and no more than 20 % B1b (C₄₇H₇₂O₁₄). The declared weight, 875.10, is that of the B1a component alone. Showing its structure would pass off the majority homologue as the whole drug.

- **Artemether/Lumefantrine (Coartem)** is a combination of two distinct active ingredients.

In neither case is a structure missing: there is no single one to show.

Group	Entries	Nature
D-01 — not single molecules	2	ivermectin (mixture of B1a/B1b homologues) and artemether/lumefantrine (combination of two active ingredients). It was six: the other four were closed with the structure from ChEMBL, see §2.4-bis.
D-02 — not representable in SMILES	19	monoclonal antibodies, proteins and peptides (trastuzumab, pembrolizumab, insulin, semaglutide, ciclosporin A...). The absence is correct, not a gap.

The choice is deliberate: a wrong datum replaced by another wrong datum is no progress, and a declared absence is preferable to an unverified presence.

3. Spectral prediction

3.1 Structural, not textual, recognition

Functional groups are identified with **SMARTS patterns applied to the molecular graph** through RDKit (`bsi-spettri.js`), not with regular expressions applied to the SMILES string.

The difference is not stylistic. The earlier textual method had two structural defects:

1. **Exclusivity.** The checks were organised in `else if` chains, so ****only one functional group per molecule**** was recognised.

1. **Blindness to structure.** A regex reads characters, not bonds.

The emblematic case is **acetylsalicylic acid** (CC(=O)Oc1ccccc1C(=O)O), which contains both an ester and a carboxylic acid. Textual recognition identified only the ester; the resulting IR spectrum lacked the broad O-H band at ~3000 cm⁻¹ and the acid C=O at ~1690 cm⁻¹ — the two bands that identify the molecule — and the ¹H NMR spectrum lacked the carboxylic proton signal at 11.6 ppm.

Graph-based recognition is **additive by construction**: a molecule may present every group it possesses.

3.2 Representation conventions

Axis conventions are not uniform across techniques, and are verified automatically (`test_assi.js` , `test_assi_canvas.js`) by extracting the tick labels from the drawing produced:

Technique	Abscissa	Direction	Ordinate
IR	wavenumber (cm^{-1})	decreasing, $4000 \rightarrow 400$	% transmittance , bands pointing down
^1H NMR	δ (ppm)	decreasing, $14 \rightarrow 0$	intensity, TMS reference at 0
UV-Vis	λ (nm)	increasing, $200 \rightarrow 400$	absorbance
MS	m/z	increasing	relative abundance

3.3 Determinism

No spectrum contains random components. Earlier versions added pseudo-random noise to the trace on every redraw, with the consequence that the same molecule produced a different spectrum each time and two spectra were not comparable. A computed profile must be **reproducible**: where a realistic baseline is needed, a deterministic function of the wavenumber is used.

Likewise, in the NMR predictor the peak heights were generated at random. In a ^1H spectrum the height **is** the quantitative information — the integration — and inventing it is worse than omitting it. Currently:

- where the number of protons is known, the height is the real integration and the label reports it (e.g. `7.26 (4H)`);

- where the prediction is a **range** (e.g. "6.5–8.5 ppm"), the band of the range is drawn with a mark on the typical value and a uniform height: the chart shows *where* the signal falls without asserting how intense it is.

3.4 Chemical effects modelled

Effect	Implementation
Carbonyl conjugation	a $\sim 25 \text{ cm}^{-1}$ lowering applied to the individual carbonyl , not globally
Phenolic ester	raising to $\sim 1760 \text{ cm}^{-1}$
C-H intensity	proportional to the number of CH_2/CH_3 groups present
Aromatic ring substitution	"out of plane" bands $900\text{--}690 \text{ cm}^{-1}$ distinguished for ortho/meta/para/mono
Aldehyde Fermi doublet	bands at 2820 and 2720 cm^{-1}
Amide bands I and II	1655 and 1550 cm^{-1}

Verification: acetylsalicylic acid produces an aryl ester at 1760 cm^{-1} and a conjugated acid at 1690 cm^{-1} — the two values that identify it — while acetophenone (conjugated ketone, 1690) and ethyl acetate (unconjugated ester,

1. confirm that the effect is applied selectively and not indiscriminately.

3.5 Declared limits

The spectra produced are **predictions based on functional-group tables**, not measured spectra nor quantum-mechanical simulations. They indicate where the bands of the groups present fall, with tabulated positions and intensities. **They do not replace an experimental database** (SDBS, NIST) nor an instrumental measurement, and the chart says so.

In particular the following are not modelled: second-order spin-spin couplings, solvent and concentration effects, combination bands and overtones, solid-state polymorphism.

3-bis. Cheminformatics and QSAR models

3-bis.1 Where the methods come from

None of the methods in the **Cheminformatics** section was invented for the occasion. They are the ones in use, with the references that define them.

Method	Reference	How it is implemented here
Tanimoto coefficient	Rogers & Tanimoto, 1960	Bits in common over bits in union, with the $0/0 = 1$ convention declared in the code
Morgan / ECFP fingerprints	Rogers & Hahn, <i>J. Chem. Inf. Model.</i> 50 (2010) 742	From RDKit MinimalLib, radius 2, 2048 bits
MACCS keys	Durant <i>et al.</i> , <i>J. Chem. Inf. Comput. Sci.</i> 42 (2002) 1273	From RDKit MinimalLib, 166 keys
Rule of five	Lipinski <i>et al.</i> , <i>Adv. Drug Deliv. Rev.</i> 23 (1997) 3	Over the RDKit descriptors, with the violation count
Veber filter	Veber <i>et al.</i> , <i>J. Med. Chem.</i> 45 (2002) 2615	Rotatable bonds ≤ 10 , TPSA $\leq 140\text{ \AA}^2$
QED	Bickerton <i>et al.</i> , <i>Nat. Chem.</i> 4 (2012) 90	Recomputed: MinimalLib exposes 43 descriptors but not QED. The desirability-function parameters are those of the paper
Butina clustering	Butina, <i>J. Chem. Inf. Comput. Sci.</i> 39 (1999) 747	Threshold on the Tanimoto distance, leaders chosen by neighbour count
Bemis-Murcko scaffolds	Bemis & Murcko, <i>J. Med. Chem.</i> 39 (1996) 2887	Ring systems plus the bonds that connect them
PAINS	Baell & Holloway, <i>J. Med. Chem.</i> 53 (2010) 2719	A subset of the SMARTS patterns, with the reason for the flag

Method	Reference	How it is implemented here
Brenk alerts	Brenk <i>et al.</i> , <i>ChemMedChem</i> 3 (2008) 435	Likewise
SALI	Guha & Van Drie, <i>J. Chem. Inf. Model.</i> 48 (2008) 646	$\Delta\text{activity} / (1 - \text{Tanimoto})$, over the pairs above the similarity threshold

3-bis.2 The four things that make a QSAR honest

A QSAR model is extremely easy to make look good. The four precautions below are what separate a number from a measurement, and all four are verified by the `test_cheminfo` bench.

Scaffold split. With a random split, close analogues from the same series end up on both sides: the model recovers what it has already seen. The scaffold split groups the molecules by Bemis–Murcko scaffold and assigns **whole scaffolds** to one side only. The R^2 that comes out is lower — and it is the one that survives the new molecule.

Null model by scrambling. The same model is retrained eight times on shuffled labels. If the true score does not exceed the best of the twins, the model has learnt nothing transferable, and the panel says so in those words. It is the check most often missing: without it, an R^2 of 0.4 over thirty molecules is indistinguishable from chance.

Applicability domain. A prediction on a molecule far from everything the model has seen is an extrapolation, not a prediction. The distance to the nearest training neighbour is shown next to every predicted value.

Scaffold-grouped cross-validation. A single split, over a small set, is a noisy number: on 28 molecules at 25 % the test set holds seven, and moving one shifts the R^2 by tenths. The folds group by scaffold — no scaffold sits in two folds — and the result is given as **mean $\pm$ deviation**, not as a single figure.

How much this matters, measured on the 28-inhibitor example. The single split gives R^2 **0.861**. Five-fold validation gives **0.605 $\pm$ 0.452**, with the individual folds at 0.861 · 0.538 · 0.795 · **−0.150** · 0.980. The single split had drawn the lucky fold. Together the two figures say what neither says alone: the model works, but it depends appreciably on which molecules it gets for training — and on a set of twenty-eight molecules that is exactly what one should expect.

3-bis.3 Declared limits

Limit	Consequence
The example sets are small (28 and 40 molecules)	They exist to show the method, not to produce a usable model. On sets that size, the null model is the only serious defence — which is why it is there
The model is linear in the kernel (kernel ridge, Tanimoto kernel) or logistic	No neural network, no gradient boosting: on a few hundred molecules these give no demonstrable advantage, and the cost would be a model one cannot inspect

Limit	Consequence
PAINS are not a verdict	A PAINS pattern signals that the compound has often been a false positive in fluorescence assays, not that it is inactive. The panel writes this next to every flag
The PCA is on the descriptors , not on the fingerprints	On binary fingerprints PCA is not very informative; loadings on descriptors can be read, and are shown
There is no large-scale substructure search	The workbench is sized for the sets one pastes in by hand, not for a library of millions of compounds

3-bis.4 Two surfaces, two roles — and a number that was false

The application has **two** places where cheminformatics happens, and they serve different purposes. The distinction is written here because an assessor who opens the wrong page first would form the wrong impression.

	rdkit_lab.html — "RDKit Lab"	Cheminformatics section
Who it is for	Someone studying: one molecule at a time, to see what RDKit can say about it	Someone working: a set of molecules, to derive a defensible model
Input	One SMILES	A pasted set, with the measured activity
QSAR	Empirical estimates from the descriptors, declared as such in the panel	A model trained on the supplied data, with scaffold split, cross-validation and null model
Output	What is on screen	CSV and method report

A number that was false, and how the bench caught it. The lab's "Pharma Pro" panel compared the molecule against eight reference drugs and displayed a similarity percentage. That percentage did not come from a fingerprint: it came from **eight bits of thresholded descriptors** — "has aromatic rings", "HBA > 4", "weight between 200 and 500" — over which a Tanimoto was computed. Measured: **caffeine against metformin gave 0.75**, while on Morgan fingerprints it is **0.024**. Two molecules with no fragments in common, shown at 75 %, on a page called "RDKit Lab" where RDKit was already loaded. Alongside that, three of the eight reference fingerprints were **written by hand** and had the "aromatic" bit wrong (caffeine, which has two aromatic rings, morphine and amoxicillin which have one), and the **SMILES for omeprazole was not omeprazole**: C0c1ccc2[nH]c(=S)cc2c10C is a structure RDKit rejects.

All corrected: the comparison uses the Morgan fingerprint with the shared engine's Tanimoto, the fingerprints are computed rather than typed in, and omeprazole is omeprazole (monoisotopic mass 345.11, three aromatic rings — both verified). The two functions of the fake similarity were **removed**, not bypassed: leaving them around would have put them back in use at the first edit, and the bench checks that they no longer exist.

One implementation of Tanimoto. The lab had its own, which agreed with the engine's on ordinary cases but answered 0 where the engine answers 1 (two empty fingerprints). The page now loads `bsi-cheminfo.js` and calls that: `test_cheminfo` compares the two routes over

all 28 reference pairs and demands a deviation of **exactly zero**.

3-ter. Alternative text for figures is not invented

The biochemistry guide carries **98 figures**. On replacing it, the accessibility bench rejected it: none had an `alt` attribute.

The easy way out was `alt=""`, and the bench would have accepted it — the comment in its code says, correctly, that for a **decorative** image this is the right thing: it tells the screen reader to skip it. But these are biochemistry figures, that is, information. Marking them decorative would have passed the check by telling screen-reader users to ignore 98 diagrams: a hollow green, of the same family as the "contrast 0" that did not cover the new panels (§3-bis) and the 100 % coverage that summed the wrong ranges.

The honest way was available: every image sits inside a `<figure>` with its own `<figcaption>`, and **the captions were written by the document's author**.

Figures	62, holding 98 images
Single-image figures	26 — the alt is the caption, minus the "Figura N." label
Two-panel figures	36
Pair captions split between the two panels	33 out of 36
Pairs sharing one caption	3 — they do not split unambiguously

All 36 pair captions distinguish the two panels with "a sinistra" (left) and "a destra" (right): where the split is unambiguous, each image receives its own half of the sentence. Where it is not, the two images share the whole caption — verbose to listen to, but **it cannot be wrong**, and a wrong description of a figure is worse than a verbose one.

Why not attempt a more aggressive split. The remaining 3 could have been squeezed out with looser regular expressions. A split that goes wrong produces "this image shows X" next to an image that shows Y, and the screen reader reads it with the same confidence as the correct ones. The fallback is noisy; the error is invisible.

Measured after the change: **98 out of 98 with alternative text, none empty**, contrast **0** over 1,897 text elements.

3-quater. From prototype to working tool

The functions described in §3-bis were enough to say something defensible about a set of molecules. They were not enough to **work** with it: on day one, a cheminformatics group asks to search a substructure across the whole set, to see the SAR table, and to know what happens to activity when a chlorine is replaced by a methyl.

3-quater.1 What the library had, and what it does not

Before designing anything, MinimalLib 2025.03.4 was **probed at runtime** rather than assumed. Probing first changed the design: three of the capabilities below had gone unused because nobody knew they were there.

Present	What it is for
<code>SubstructLibrary</code>	Substructure search with a pattern fingerprint: it discards in bulk the molecules that cannot match, instead of attempting isomorphism on each
<code>get_rxn</code> · <code>Reaction.run_reactants</code>	Chemical transformations. This is what makes it possible to cut a bond, and from there come matched pairs and the SAR table
<code>generate_aligned_coords</code>	Aligning molecules to a common core
<code>get_molblock</code> · <code>get_v3Kmolblock</code>	SDF export, the format in which sets are exchanged between groups and programs

Absent	Declared consequence
<code>cleanup</code> , <code>neutralize</code>	Standardisation stops at the largest fragment and the canonical SMILES: charges are not neutralised
<code>canonical_tautomer</code>	Two tautomers written differently remain two distinct entries
<code>FragmentOnBonds</code> , <code>RWMol</code>	No direct molecule editing: cutting goes through a reaction
Maximum common substructure (MCS)	A SAR table's core must be supplied , not deduced. The panel offers eight ready cores and a free field

3-quater.2 The cut, and why two rules are needed

A reaction SMARTS breaks a bond and marks both ends with a dummy atom. Verified at runtime: CCOc1ccccc1 gives three cuts under the fine rule (CH3-CH2, CH2-O, O-aryl) and two under the coarse one; benzene and methane give none.

Rule	Cuts	When it is needed
fine (default)	Any acyclic single bond, including terminal substituents Cl, CH ₃ , OH, F	SAR table and fine transformations (<chem>Cl</chem> → <chem>CH3</chem>)
coarse (BRICS-style)	Excludes terminal atoms	Large sets, where fine cuts are too many; produces larger variable parts (" <chem>chlorophenyl</chem> → <chem>phenyl</chem> ")

The fine rule was born of a defect. The first draft used only the coarse one, and the SAR table of the test series **lost the chlorine on the ring**: it found the substituent on the nitrogen and missed exactly the column that matters, because chlorine is terminal. The fine rule is a superset of the coarse one: every cut the coarse rule finds, the fine one finds too.

3-quater.3 Matched molecular pairs

Two molecules form a pair when, cutting one bond in each, they are left with the **same context** and two different variable parts. The activity difference is attributed to that substitution and to nothing else — which is why this reading beats a correlation over the whole set, which averages over molecules differing in ten ways.

Verified on a series **built with known effects**: if the analysis does not recover them, it is not usable on real data.

Transformation	Expected	Found	Concordant pairs
chlorophenyl → phenyl	−1.0	−1.0	2 of 2
methylphenyl → phenyl	−0.5	−0.5	2 of 2
methyl → chlorine	+0.5	+0.5	2 of 2
ethylamide → propylamide	+0.2	+0.2	3 of 3

On the 28 example inhibitors, taken from ChEMBL: **449 pairs, 98 transformations** seen at least twice, 199 seen once, in 133 ms. At the top, **CF₃ → SO₂NH₂ with a median Δ of −2.65** over two concordant pairs.

Two declared choices:

- the **direction is normalised** on the alphabetical order of the two parts.

Without it the same effect would appear twice, with opposite medians, and neither would carry the right pair count;

- transformations seen **fewer than twice** are excluded from the table and

counted separately. A transformation seen once with Δ = +3 is not a discovery: it is an anecdote, and putting it at the top would have decisions made on it.

The "concordant" column is the one that decides whether to trust it. A median of +1.0 over three pairs one of which is −2.0 is not the same as +1.0 over three pairs all positive, and the median alone does not say so.

3-quater.4 The SAR table, and how the position is found

The hard part is not finding the substituents: it is knowing which **position** each occupies. With no MCS and no molecule editing, the solution exploits a property of the cut: **the fragment containing the core carries the dummy atom exactly at the attachment point**. Matching the core onto that fragment reveals which of its atoms the dummy hangs off, and therefore which position the substituent occupies.

This also yields a necessary and free filter: if the dummy is **not** attached to a core atom, the cut happened inside a substituent and that piece is not the whole substituent. It is discarded.

On the 28 inhibitors, with a benzamide core: **20 molecules with the core, 2 positions, 8 excluded**. The excluded ones do not appear in the table: showing them with empty cells would make them look like part of the series with substituents that were not found.

A naming defect, and how it showed itself. `scaffoldMurcko()` returned its result in a field called `smiles`, but that value **is not a SMILES**: it is a canonical fingerprint of the scaffold, built so that two molecules with the same scaffold produce the same string. It serves to **group** — folds, scaffold split — not to query. The SAR panel took it for a SMILES and proposed `6,6,6,6,7,...|0-13:1,...` as the core; decomposition answered "core not interpretable".

The field is now called `chiave` (key) and declares `eUnoSmiles: false`; `smiles` remains as an alias. The bench pins that equal scaffolds give the same key, that different scaffolds give different keys, and that the key **does not read back as a SMILES**. A field's wrong name is a defect like any other.

3-querter.5 The rigour an assessor demands

Five things a modelling group puts in every report, and whose absence is the first objection in a meeting.

	Method	Why
Model comparison	Trivial reference (predict the mean), kNN over Tanimoto, kernel ridge — all on the same folds , grouped by scaffold	Comparing a model evaluated on one split with another evaluated on a different split is not a comparison. And the trivial reference sits at the top of the list, because it is the number every other must beat
Nested validation	Hyperparameters (k, lambda) are chosen on folds internal to the training set	Trying ten of them on the test set and reporting the best inflates the score: that number is not a prediction, it is the maximum of ten attempts
Conformal intervals	Split-conformal, quantile $[(n+1)(1-\alpha)]$ of the absolute residuals on a calibration portion split by scaffold	"7.2" and " 7.2 ± 0.3 " are two different pieces of information, and only the second says whether synthesis is worth it. Calibrating on analogues of the training set would give intervals too narrow precisely on new molecules
Enrichment	EF at 1 %, 5 %, 10 % with the maximum achievable alongside; BEDROC (Truchon & Bayly, <i>J. Chem. Inf. Model.</i> 47 (2007) 488)	In a screen one does not look at the whole list. A ROC-AUC of 0.80 can hide a top-of-list with not a single active, because AUC also rewards the ordering in the tail, which nobody will buy
Learning curve	Score against training-set size, with the test set whole at every point	It answers the only question that matters when the model is mediocre: more chemistry or more data?

The comparison's verdict is proven in both directions, like the null model. On a real signal — activity proportional to halogen count, over ten distinct cores — kernel ridge reaches R^2 0.943 with a margin of 0.943 against a dispersion of 0.035, and the verdict is *beats it*. On the **same molecules with random labels** the best stops at -0.265 , margin -0.034 , and the verdict is *does not beat it*. A guard that never fires and one that always fires are the same defect.

Enrichment is verified on the three cases where the right value is computable by hand:

Ordering (2 actives of 10)	AUC	EF@20 %	BEDROC
Perfect: both actives on top	1	5 — and it is the maximum achievable	1
Worst: both actives last	0	0	0, not a negative
Actives at ranks 2 and 5	0.75 = 12/16	2.5 = 1/0.4	0.119

The worst case is the one that counts: without it one would not know whether BEDROC reaches zero or goes below, and a negative BEDROC is the sign that the formula's additive term is missing.

3-quater.6 Limits of this block

Limit	Consequence
Pairs grow with the square of the molecule count	Measured: 6 molecules 45 ms, 40 molecules 227 ms with 1,255 pairs. Beyond 300 molecules the panel refuses to start and says so, instead of freezing the page
The cut is single	A pair differing by two substitutions at once is not found. This is the limit of the single-cut scheme, shared with the standard tools
The same transformation appears in several forms	Cuts at different positions give different contexts for the same chemical change (<code>*CC>>*CCC</code> , <code>*NCC>>*NCCC</code> , <code>*C(=O)NCC>>*C(=O)NCCC</code>). This is expected redundancy in the single-cut scheme, not an error
Charges are not neutralised and tautomers not canonicalised	MinimalLib does not allow it (§3-quater.1). A set mixing neutral and ionic forms of the same molecule counts them as distinct entries
The SAR table's core must be supplied	No MCS available. The panel proposes nothing automatically, because a wrong proposal is worse than no proposal

4. Physical constants and tabulated data

The lookup of physical constants proceeds by decreasing specificity (exact match → symbol → significant words → substring of at least 4 characters) and **collects every candidate**: if more than one remains, the request is refused with the list of possibilities rather than arbitrarily returning the first. A wrong physical value returned with confidence is worse than a request for clarification.

The exam molecules (`MOLECOLE_ESAME`) are tabulated with verified values; the recognition of groups from SMILES uses a `consume` field, so that each group removes the portion of structure it has explained and no overlapping counts are produced.

5. Automation of the verification

All the checks described are runnable and repeatable. At the time of writing the battery includes, among others:

Bench	Subject	Checks
<code>tools/verifica-farmaci.js</code>	structure ↔ molecular weight, duplicates, recorded deviations	233 entries
<code>test_spettri</code>	group recognition on reference molecules	36
<code>test_assi /</code> <code>test_assi_canvas</code>	axis conventions (SVG and canvas)	19
<code>test_costanti</code>	physical-constant lookup and refusal of ambiguities	45
<code>audit_dati</code>	consistency of the tabulated data	29
<code>verifica_guida</code>	correspondence between what the documentation promises and what the code does	71

The last bench deserves a mention: it verifies that the statements contained in the user documentation correspond to the real behaviour of the code. A promise the documentation does not keep is a defect on a par with a computational error, and is treated as such.

6. Transparency statement

This document describes checks that are **actually implemented and runnable**, with the results actually obtained and the limits of the predictors. At version `bsi-v181` the residual errors on the drug database are **zero**: the 21 deviations that remain are entries without a structure, each with its own recorded reason, not errors passed over in silence.

Where a datum is not verifiable with the instruments available, that is declared rather than presented as verified.

Document updated to version `bsi-v181`.

Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	bsi-v181
Commit	e19afc6eb78c861aed3b485e13cc5c3fc904af8f
Generated (UTC)	2026-10-02T16:27:01.565Z
Machine-readable format	<code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	<code>RDKit_minimal.js</code>	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	<code>RDKit_minimal.wasm</code>	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	<code>three.min.js</code>	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	<code>three_bloom.js</code>	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	<code>gltf_loader.js</code>	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	<code>3Dmol-min.js</code>	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).
SmilesDrawer _{Daniel Probst}	<code>smiles-drawer.min.js</code>	MIT	246 kB	2D structure drawing from SMILES.
sql.js (JS glue) _{sql.js / SQLite}	<code>lib/sql-wasm.js</code>	MIT	45 kB	In-browser SQLite database for the data exercises.

Component	Version/File	Licence	Size	Role
SQLite (WebAssembly) _{SQLite Consortium}	lib/sql-wasm.wasm	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	lib/dimuon.js	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
RDKit_minimal.js	58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9
RDKit_minimal.wasm	e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
glTF_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d

2. Code written for this project

File	Size	Role
index.html	4657 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	33 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	114 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
sw.js	5 kB	Service Worker: offline operation and network strategy.
rdkit_lab.html	260 kB	Cheminformatics laboratory.
astro.html	2660 kB	Astrochemistry module.
chimorga.html	194 kB	Organic chemistry module.
proxy/spectra-proxy.js	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
index.html	6232410e66315da479e771bbc481c23b3c9d79853deb424617e5d1cb98650305
bsi-ai-hub.js	985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476
bsi-spettri.js	1eaf605c2d93f00c3edaa843de504c8d6625131b75d8bb4a0bef8d9b767b49bd
bsi-cheminfo.js	33cd76132c4c3d495b5553315c15a44e991f56a8f1847104f2ec198aa0c005fc
sw.js	ce8c1005668331f3d7b8f0d6b8701c6aad9805ce091e040ca14c0fe45a39a219
rdkit_lab.html	066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad
astro.html	3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d
chimorga.html	047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d
proxy/spectra-proxy.js	808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
playwright-core	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

The application's licence is distinct from those of its components. The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see [LICENSE](#)): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

Generated on 2026-10-02T16:27:01.565Z.

Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	bsi-v181
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

How to read this matrix. Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#) , which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	Structural data for drugs must be consistent with the declared molecular weight	FARM_DATA in index.html	tools/verifica-farmaci.js — RDKit ↔ literature comparison over 233 entries
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	bsi-spettri.js §1, SMARTS patterns via RDKit	test_spettri — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	bandeDaGruppi() , additive construction	test_spettri §1 — acetylsalicylic acid: ester and acid
SCI-04	Spectra must respect the conventions of their technique	svgIR() , makeNMRSvg() , drawUVSpectrum() , drawMSSpectrum()	test_assi (SVG), test_assi_canvas (canvas)
SCI-05	A computed spectrum must be reproducible	every Math.random() removed from the traces	test_assi §3, test_spettri §8 — byte-for-byte comparison
SCI-06	The height of ¹ H peaks must represent the integration, or not be asserted at all	makeNMRSvg() ; _nmrSvg() draws bands of uniform height	test_assi §4

ID	Requirement	Implementation	Verification bench
SCI-07	Conjugation must act on the individual carbonyl	SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code>	<code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	<code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate	<code>test_costanti</code> — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; <code>docs/06</code> §3.5	<code>verifica_guida</code> — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	<code>ssimm</code> section in <code>index.html</code>	<code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	<code>scroma</code> section: $\sigma = t_{R}/\sqrt{N}$, R_s from distance and widths	<code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	<code>staratura</code> section: ordinary least squares, F-test line↔parabola, deleted residual	Verified on the three example sets included: linear ($F = 0.3$), non-linear ($F = 186.6 > 7.71$), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	<code>bsi-cheminfo.js</code> §9 — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code>	<code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test
SCI-15	A model score must be compared against chance, not presented on its own	<code>bsi-cheminfo.js</code> §9 — retraining on scrambled labels, eight repetitions	<code>test_cheminfo</code> — verified in both directions : on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 -0.199 does not beat them
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	<code>bsi-cheminfo.js</code> §1 — deduplication on the canonical SMILES after the largest fragment	<code>test_cheminfo</code> — <code>OC(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	<code>bsi-cheminfo.js</code> §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	<code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	<code>bsi-cheminfo.js</code> §9-bis — <code>validazioneIncrociata()</code> over scaffold-grouped folds	<code>test_cheminfo</code> — the folds cover every molecule once, no scaffold sits in two folds , and the same seed gives the same folds

ID	Requirement	Implementation	Verification bench
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	test_cheminfo — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	bsi-cheminfo.js §9-ter — esportaCsv() and rapportoMetodo(), both formed in memory in the browser	test_cheminfo — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	rdkit_lab.html loads bsi-cheminfo.js and uses its tanimoto() and fingerprint(); the eight-bit similarity has been removed	test_cheminfo — page and engine give the same Tanimoto over all 28 pairs , deviation exactly 0 ; the fake-similarity functions no longer exist; every reference structure is valid and carries the monoisotopic mass of the drug it names

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	modelloSuggestitoDaErrore(), bsi_modelli_ko blacklist	test_404 — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	bsi_prov_ko, 24 h TTL, providerUtilizzabili()	test_ko (35), browser_ko (37)
AI-03	A temporary overload must produce a retry, not an error	erroreTemporaneo(), TENTATIVI_TEMPORANEO	test_503 — 40 checks
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	ATTESA_TROPPO_LUNGA_MS	test_attesalunga — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	tettoDaErrore(), MARGINE_TETTO	test_tetto — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in appendAgentTurn()	test_firma — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	_turnoInCorso in send()	test_doppioinvio — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	provaTuttiIFornitori()	browser_prova — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by PROVIDERS[id].free	browser_prova §D2

ID	Requirement	Implementation	Verification bench
AI-10	A documented route must exist to remove CORS and rate limits	<code>proxy/spectra-proxy.js</code> + in-app guide	<code>test_guidaproxy</code> (12), <code>browser_proxy</code> (15), <code>browser_proxyui</code> (17)

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	<code>audit_stabilita</code> §1 — 89 sections × 5 rounds
STA-02	Exhausting <code>localStorage</code> must not make the app unusable	guarded writes; persistent warning in the File Manager	<code>audit_quota</code> (10), <code>test_filemanager</code> (15)
STA-03	No rejected promise may remain unhandled	systematic <code>.catch()</code>	<code>audit_promesse</code> — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	<code>loadJSON()</code> with fallback	<code>audit_stabilita</code> §4
STA-05	A slow network must not be treated as an absent one	<code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>	<code>test_sw</code> §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	<code>BSI_SW_UPDATED</code> message, decision taken page-side	<code>test_sw</code> §4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	<code>test_visore3d</code> — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	<code>dbStore()</code> over a promise; persistent warning	<code>test_filemanager</code> §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	<code>audit_stabilita</code> §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	<code>bsiNitido()</code> on the canvas contexts	<code>audit_grafici</code> — 40 canvases
UI-03	The app must work at 390 px width	<code>auto-fit</code> grids, no fixed column	<code>audit_mobile</code> — the document's <code>scrollWidth</code> across all 89 sections at 390 px; <code>audit_stabilita</code> for errors in a phone viewport

ID	Requirement	Implementation	Verification bench
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	test_aggiorna — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	bsiCancellaDati() by group	browser_reset — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	test_cheminfo — the analysis is run, the panels opened in turn, 460 text elements measured, 0 defects , with the WCAG formulas rewritten inside the bench
UI-09	Opening a section must not block the page	3Dmol's first render() moved out of the click; the 296 synthesis figures drawn in 8 ms slices	test_fluidita — 89 sections traversed , median switch 14 ms, worst 162 ms, 0 sections above 100 ms ; plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in localStorage or in the Worker's secrets	tools/verifica-sicurezza.js — 235 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	audit_rete — direct verification : a canary value seeded into 71 stores of the user's data, the application used across 89 sections on 6 pages, and the URL, headers and body of every request inspected. Plus verifica-sicurezza on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in file_manager.html	tools/verifica-sicurezza.js
SEC-04	No conflict marker may reach publication	—	verifica_guida §12 — 58 text files

SEC-03 — declared limitation. GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v181`. The old one remains in the history and no longer opens anything.

6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across 13 pages and 89 sections (<code>tools/verifica-accessibilita.js</code> , 33,642 text elements: it was 19,751 before text over gradients entered the measurement). Of the 1,069 contrast defects that emerged, none remain: 0 measured over those same 89 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are zero : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (<code>docs/02</code> §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	not representable : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4
PERF-02	Code coverage of the verification benches	partially measured : 49.89 % of statements over the widest path (<code>audit_copertura</code>). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	not verified — the battery runs on Chromium only; see §8

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01...21)	20	20	100 %
AI agent (AI-01...10)	10	10	100 %
Stability (STA-01...08)	8	8	100 %
Interface (UI-01...09)	7	7	100 %

Category	Requirements	Verified by a bench	Coverage
Security (SEC-01...04)	4	4	100 %
Automated total	49	49	100 %
Not automated (§6)	5	0	0 %
Declared total	54	49	91 %

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

Why two totals. Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **49 requirements verified out of 54 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: 49.89 % of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers 403 to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page

Gap	What it entails	Why it is so
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See 06-Scientific-Accuracy-Data-Provenance.md §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

_Document updated to version bsi-v181 . _

Release Conformance Statement — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version	bsi-v181
Author and release owner	Samuele Pio Provenzano
Repository	github.com/samupropiol-ship-it/BioSpecInfo-v11
Distribution	GitHub Pages — samupropiol-ship-it.github.io/BioSpecInfo-v11/
Nature of the software	Client-side Progressive Web App, no backend

1. Subject of this statement

The undersigned declares that version `bsi-v181` of BioSpecInfo has been put through the verification procedure described in `02-Verification-Validation-Report.md` and that the outcomes are those reported, without selection, in `../evidence/RAPPORTO-VERIFICA.md` — a document generated automatically by `tools/genera-evidenza.js`, which transcribes the benches' output with no manual intervention.

This statement does **not** attest the absence of defects. It attests that:

1. an automated verification procedure exists and can be reproduced by third parties;
 1. the outcome reported is the one actually produced, failures included;
 2. the known limitations are declared explicitly and not presented as features.
-

2. Intended use and limitations

2.1 Declared purpose

BioSpecInfo is a **teaching and study-support** tool for chemistry, biochemistry, pharmacology and astrochemistry. It is designed for university students and for exam preparation.

2.2 What the software is NOT intended for

These exclusions are an integral part of this statement.

Domain	Exclusion
Clinical use	It is not a medical device within the meaning of Regulation (EU) 2017/745. It does not support diagnosis, prescription, dosing or therapeutic decisions. The pharmacological data are for teaching purposes only, and the application says so at the head of the section.
Certified analytical use	The spectra are predictions from functional-group tables, not measurements nor quantum-mechanical simulations. They do not replace an experimental database (SDBS, NIST) nor an instrumental determination, and cannot be used for the formal identification of substances.
Regulated GxP environment	It is not validated for use in a GLP/GMP environment. The alignment with GAMP 5 described in <code>docs/03</code> is methodological, not a formal qualification.
Access control	The File Manager protects its contents with a client-side deterrent. On a static site there is no effective access control: whoever reads the source bypasses it. It must not be used for confidential material.

3. Summary of verification outcomes

The full outcomes, with each bench's textual output, the execution environment, the commit and the SHA-256 digests, are in `../evidence/RAPPORT0-VERIFICA.md`.

Family	What it demonstrates
Scientific data	The chemical data are verified against an independent source
AI agent	The assistant stays usable when the external provider fails
Stability	The app withstands long sessions, exhausted storage, degraded networks
Interface	The panels respond as documented
Documentation/code consistency	What the documentation promises exists in the code

4. Known deviations at the date of release

Listed in full. None has been removed from verification in order to make it pass.

#	Deviation	Impact	Treatment
D-01	2 entries (it was 6) that are not single molecules: Ivermectin, a mixture of B1a/B1b homologues, and Coartem, a combination of two active ingredients	For these two no 2D/3D representation and no spectral prediction is available	Entries left without SMILES : the pharmacological data remain, the structure is not shown. For Digoxin, Vincristine and Tacrolimus (topical and systemic) the structure was taken from ChEMBL and passes the comparison with the declared molecular weight: those four left the registry. The two that remain have no structure to show, not a missing one.

#	Deviation	Impact	Treatment
D-02	19 entries with no SMILES by their very nature (monoclonal antibodies, peptides)	None: for those molecules SMILES notation is not the appropriate representation	Not a substantive deviation; listed for completeness
D-03	SEC-02 verified only by proxy — remedied: see S-10	—	<code>audit_rete</code> observes the traffic during real use, with a canary value seeded into the user's own data
D-04	The File Manager password in the git history — remedied: see S-12	—	The password was changed at version <code>bsi-v181</code> . The old one is still readable in the history and no longer opens anything
D-06	Code coverage measured but partial: 49.89 % of statements, not branches	It is the coverage of the widest path a bench walks (89 sections plus the other pages), not of the whole battery; and an <code>if</code> entered from one side only counts as covered	<code>audit_copertura</code> measures it with Chromium's profiler, with no build and without rewriting the source. The value is recorded: if it falls, the battery fails
D-07	Verification on Chromium only	Firefox and WebKit are tested by hand	Declared in <code>docs/08</code> §8. In the verification environment the reason is checkable: the CDN from which <code>playwright-core</code> downloads the other engines answers 403 to the network policy, so Firefox and WebKit cannot be installed there
D-08	Partial English translation — remedied: see S-13	—	All 16 documents are in English. <code>verifica-documenti</code> and <code>verifica-affermazioni</code> read <code>docs/en/</code> just like the Italian set: a disagreement between the two languages fails the battery
D-09	Full WCAG 2.1 AA conformance not verifiable entirely by automation	What stays outside is text inside SVGs (8,888 elements), text over a real background image (108) and everything requiring human judgement. Text over a gradient has entered the measurement: 591 elements, judged against the worst stop of the gradient	The skipped elements are counted and reported on every run

Contrast, from 1,069 defects to zero — how it surfaced and how it was closed

Up to version bsi-v168 this document declared WCAG conformance "verified by sampling" and the bench reported **0 defects**. Those were two statements at once true and meaningless: the bench inspected only the **visible** elements, and index.html shows one section at a time. Out of 19,751 text elements it was looking at **41**.

Walking all 87 sections then present, **1,069** contrast defects surfaced. They had not appeared: they had always been there.

Today they are **zero**, measured with the same bench over every section — 88 of them now. Not by wholesale substitution — that road, attempted once, introduced 997 new defects — but by going back to the **common cause** every time and re-measuring after each change. The **40 unlabelled fields** have likewise fallen to **zero**.

The gravest causes were not wrong colours but **theme tokens whose value equalled the background**:

Structural defect	Measured effect
--g800 is #0d1522 , identical to --bg — and the rule a { color: var(--g800) } applied it to every link	Every link without a colour of its own was at contrast 1:1 : invisible
--g700 is #16263d , nearly identical to --white (#16273e)	Text at 1.01:1
Buttons that changed their background without changing their text colour	The selected state (or the deselected one, depending on the panel) became unreadable: 1.39:1
lightenColor(hex,pct){ return hex; }	A function called "lighten" that returned the colour unchanged : an unfinished stub, in all likelihood written for this very problem
The quiz answer panel: light background, light text	The student could not read the correct answer

The structural remedy was an explicit token — --testo-forte — for prominent text on a dark surface, applied only to the uses as a *text colour*: --g800 and --g700 stay where they serve as background or border.

Cause	Defects	Remedy
A dark-theme safety net with higher specificity than the components: text meant for the white cards was forced to #d4dce6 , contrast 1.38:1	159	Rewritten with :where() (zero specificity): it protects text that has no colour of its own and stops fighting text that has one
A per-category accent used as text on a dark header, with #0d1522 (near-black) as fallback: 1.13:1	~400	bsiAccentoLeggibile() preserves the hue and raises the lightness just enough. Applied at the 14 points where the accent is chosen — not in the data, so it holds for future entries too

Cause	Defects	Remedy
Two hues used only on dark backgrounds, just under threshold (3.75:1 and 3.04:1)	139	Replaced after verifying that they never appeared on a light background
Table zebra striping white/near-black with always-light text: the white rows were unreadable	57	Two dark tones, consistent with the theme
Periodic-table cells just under threshold (4.46:1 and 3.88:1)	43	Text at #e4e4e4, verified across all nine categories
"#var(--text)" — one # too many makes the colour invalid: in SVG it falls back to black, on canvas the assignment is ignored and the previous colour persists	8 places	Replaced with the real value

The direction of the correction depends on the surface, not on the colour. The first attempt always lightened: it removed thirty defects on the dark header and added eighteen on the light formula box that uses the same accent. The function now darkens on light backgrounds and lightens on dark ones.

The final tail: 80 → 0. The last eighty were scattered over 33 distinct causes and looked like finishing work. They were not: three more structural causes lay underneath.

Final cause	Remedy
The "automatic contrast corrector" was producing defects instead of removing them. It used <i>perceived</i> brightness instead of WCAG luminance, so its thresholds corresponded to no real ratio; it ignored alpha , so <code>rgba(255,180,84,.08)</code> over a dark surface looked to it like a "light background" and it darkened the text to #16273e — creating a defect at 1.21:1 on top of a source that was already correct; and it looked only at elements carrying a background of their own , that is, never the commonest case	Rewritten: it walks the ancestors compositing alpha to obtain the effective background, measures the true WCAG ratio, and delegates the correction to <code>bsiAccentoLeggibile()</code>
--g900 is #0d1522 — the same as --bg , exactly like --g800: the legacy of a greyscale born for the <i>light</i> theme, where --g900 was "the darkest text". With the theme inverted, that role no longer exists	The uses as a text colour (30 rules today) move to --testo-forte; the 16 uses as a background stay where they are
A colour chosen from data used as a background with the text hard-coded to #fff: it works while the hue is dark and stops as soon as someone adds #e65100	<code>bsiEtichettaLeggibile()</code> — the dual of the previous function: it picks the text by looking at the background and, if neither extreme suffices, darkens the background while preserving the hue , because a label must stay recognisable by colour

The steps as measured with the official bench, not estimated: **80 → 27 → 26 → 10 → 4 → 0**. Each step is a full re-measurement over the 87 sections after a group of changes; none of the figures above is attributed to a single cause, because the causes were not

measured one by one.

The number remains **recorded as a baseline** in `evidence/accessibilita-riferimento.json`. Zero is not an achievement to file away: it is the value the bench defends. If a future change reintroduces even one defect, the battery **fails**.

What stays outside automation is what automation cannot judge: text inside SVGs and text over gradient backgrounds (D-09). Those elements are **counted** and reported on every run, not passed over in silence.

4-bis. Deviations remedied in this version

They are listed because they concerned **the verification apparatus itself**, and a conformance statement that says nothing about the defects of its own evidence is not worth the paper it is written on.

#	Deviation remedied	Why it mattered
S-01	32 of the 39 benches were not in the repository: they lived in the working area of the session in which they had been written	It made false the sentence this whole document rests on — "anyone can re-run them". Whoever cloned the repository found 7 out of 39. They now live in <code>tools/banchi/</code>
S-02	A missing bench was counted but did not make the battery fail	A battery missing <i>all</i> its benches would have declared "CONFORMANT" with zero failures. A missing bench now makes the outcome NON-CONFORMANT
S-03	<code>package.json</code> and <code>package-lock.json</code> were excluded from the repository	<code>npm install</code> , the documented command for reproducing the evidence, installed nothing: the battery failed at the first bench with <i>Cannot find module</i>
S-04	The accessibility check inspected only the visible section	On <code>index.html</code> that meant one section out of 87. The "0 defects" held for the Dashboard, not for the application
S-05	<code>SCI-10</code> was defined twice in the matrix, and the coverage totals were wrong (36 declared out of 37 defined)	The first number an assessor reads was wrong. A bench now verifies it
S-06	50 broken links out of 127 in the delivery packages	Documentation handed to third parties with references that led nowhere
S-07	15 attachable PDFs stuck at documents 00-05 and at an earlier version	An obsolete attachment states false things with an air of authority
S-08	The numeric-claims bench looked at the Italian documents only	The English set drifted undisturbed: <code>docs/en/05</code> declared "84 sections" while the app has 87, and stayed stuck at version <code>bsi-v146</code> while the code was at 171. The English set is now compared against the measurement just like the Italian one, and a disagreement between the two languages fails the bench

#	Deviation remedied	Why it mattered
S-13	The English translation stopped at 10 documents out of 16	A non-Italian-speaking assessor read less than two thirds of the documentation, and the missing part was the operational one: data model, interfaces, deploy, functional specifications, manual. All translated: the English set is now as complete as the Italian one, and the two consistency benches check it the same way
S-12	The File Manager password had been left in clear text in the git history	Taking it out of the files does not take it out of the history: <code>git log -p</code> hands it to anyone, and as long as that password is in use the deterrent deters nobody. The only remedy is to change it , and that has been done: the source carries only the new SHA-256 digest, the old password remains in the history and no longer opens anything. Verified in the browser: the new one opens, the old one does not
S-10	SEC-02 — "no personal data leaves the device" — was verified by checking the two <i>mechanisms</i> of exit, not the exit	A check on mechanisms says "I cannot see how it would leave", which is not "it did not leave". <code>audit_rete</code> seeds an unrepeatable value into 71 stores of the user's data, then uses the application and inspects the URL, headers and body of every request. Proved in reverse with <code>BSI_PROVA_FUGA=1</code> , which causes a leak on purpose: the bench must fail, and it does
S-11	The deviations registry had no converse guard: an entry declared that no longer deviates simply stayed	An exemption nobody revokes is a permission left switched on, and tomorrow it would silently cover a wrong structure put in its place. The bench now fails for that too — verified by putting a remedied entry back
S-09	The "automatic contrast corrector" created the defects it was meant to remove	It ignored alpha: a <code>rgba(255,180,84,.08)</code> background over a dark surface looked light to it, and it darkened with <code>!important</code> a text that was already correct in the source. A remedy tool that worsens the thing it remedies is the hardest kind of defect to see, because it presents itself as the solution

5. Acceptance criteria applied

A version is published only if **all** the following criteria are satisfied.

- ☒ No non-network JavaScript error on any page
- ☒ `tools/verifica-farmaci.js` — no defect; every deviation recorded
- ☒ `verifica_guida` — no promise in the documentation without a counterpart in the code
- ☒ No conflict marker in the tracked files
- ☒ Service Worker cache version bumped and aligned with `BSI_APP_VERSION`
- ☒ Known deviations listed in §4 of this document
- ☒ Evidence report regenerated against the published commit
- ☒ **No missing bench** — a bench that is not there is not a bench that passed
- ☒ `tools/genera-pacchetti.js` closed with **0 broken links**
- ☒ `tools/genera-pdf.js` re-run: the attachments describe this version

6. Change management

Aspect	Procedure
Traceability	Every change goes through a pull request describing the defect, its cause and its verification
Versioning	The application version coincides with the Service Worker cache version (<code>bsi-vNNN</code>), incremented at every publication
Distribution	Automatic from <code>main</code> via GitHub Pages; the Service Worker signals the new version to open sessions without interrupting them
Regression	The full battery is re-run before every merge into <code>main</code>
Reversibility	Every version corresponds to a commit; rollback is a <code>git revert</code>

7. Intellectual property and licences

All third-party components are distributed under **permissive** licences (MIT, BSD-3-Clause, CC0, Public Domain), mutually compatible and free of strong copyleft constraints. The full bill of materials, with digests and residual obligations, is in `07-SBOM.md` and in CycloneDX format at `../evidence/sbom.cdx.json`.

The code written for this project is the author's own work.

8. Data handling

The application is **local-first**: there is no backend, no personal data is collected and no persistent identifier is present. Data generated by the user (notes, chats, API keys, uploaded files) stay in the device's `localStorage` and IndexedDB and can be deleted selectively from the interface itself.

Network calls are optional and go to documented public services (§4 of `07-SBOM.md`). If the user configures the proxy, the API keys reside on a Worker they own and not on the device.

Details in `03-Security-Privacy-Compliance.md`.

9. Signature

This document is drawn up by the author of the software, who takes responsibility for it. It does not constitute certification by a third-party body, and is not presented as such.

Anyone can verify what is declared here by re-running the procedure in §5 of `../evidence/RAPPORTO-VERIFICA.md` against the commit indicated, and comparing the SHA-256 digests of the files.

Samuele Pio Provenzano _Version `bsi-v181`_

Interface Reference — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v181
Purpose	Document the project's HTTP interfaces and the external APIs the application consumes.

0. A necessary premise

BioSpecInfo exposes no API. It is an entirely client-side Progressive Web App, served as static files: there is no backend, no application server, no REST endpoints belonging to the product.

This document would therefore be empty, were it not for two interfaces that do exist and that an assessor is entitled to see documented:

	What it is	Documented in
Spectra proxy	An optional Cloudflare Worker, owned by the user, that forwards calls to the AI providers. It is the only HTTP interface written in this project.	\$1, \$2
External APIs consumed	Public third-party services the application queries, always optionally.	\$3
Internal JavaScript APIs	The extension points of the code, for whoever has to work on it.	\$4

An OpenAPI specification of non-existent endpoints would be plausible and false documentation. Only what exists is described here.

1. Spectra proxy — overview

The proxy solves two problems that have no other solution from the browser:

- CORS.** Several AI providers do not send the headers that allow a web page to call them directly. From the Worker the call originates from a server and the constraint does not apply.
- Custody of the keys.** With the proxy the API key lives among the Worker's secrets and not on the user's device.

It is **optional**: without a proxy the application works using the keys the user enters locally.

Deploy

```
cd proxy
npx wrangler login
npx wrangler deploy
npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
```

Configuration

Variable	Type	Default	Description
ORIGINI	secret/variable	(empty = all)	List of permitted origins, comma-separated. Leaving it empty makes the proxy open to anyone: not advisable.
LIMITE_IP	variable	20	Requests per single IP within the time window
TETTO_GIORNO	variable	2000	Total daily requests
GROQ_KEYS , GEMINI_KEYS , ANTHROPIC_KEYS , OPENAI_KEYS , XAI_KEYS , NVIDIA_KEYS , ZAI_KEYS , DEEPSEEK_KEYS	secrets	—	API keys. Several keys comma-separated: when the first one's quota is exhausted (429), the proxy moves to the next within the same request .

The variable names are kept in Italian because that is what they are called in the Worker: translating them here would document a configuration that does not exist.

Endpoints

Method	Path	Description
OPTIONS	/*	CORS preflight
GET	/stato	Which providers have a key configured
GET POST	/{provider}/{path...}	Forwarding to the provider

Recognised providers: anthropic · gemini · groq · xai · nvidia · zai · openai · deepseek

Status codes

Code	Meaning
200	The provider's answer, forwarded as it is
204	CORS preflight accepted
400	Request body unreadable
403	Origin not permitted by the ORIGINI list
404	Unknown provider
405	Method other than GET, POST or OPTIONS
429	Per-IP limit or daily ceiling exceeded

Code	Meaning
503	No key configured for that provider

Security choices

- **No header is forwarded blindly.** Only `anthropic-version` and `anthropic-beta` pass through; any credential sent by the client stays out. A proxy that forwards everything is an open relay.
- **The `key` parameter is not propagated.** If a client puts it in the query string, the proxy discards it: the key is the server's.
- **The body is read once and kept,** because retrying with a different key requires being able to send it again — and a stream is consumed.

2. OpenAPI 3.0 specification

The `description` fields are translated here; the **error messages in the examples are left in Italian**, because that is what the proxy actually returns to a user of the Italian application. Translating them would document responses that do not exist.

```

openapi: 3.0.3
info:
  title: Spectra Proxy
  version: "1.0.0"
  description: |
    Optional proxy (Cloudflare Worker) for calls to language-model providers.
    It removes CORS blocks and keeps the API keys on the server rather than on
    the user's device.

    It is not part of the BioSpecInfo application: it is published by the user
    on their own Cloudflare account, and the application uses it only if it
    knows its address.
  license:
    name: Proprietary – All rights reserved (Samuele Pio Provenzano)
servers:
  - url: https://{workername}.{account}.workers.dev
    description: The user's instance
    variables:
      workername: { default: spectra-proxy }
      account:    { default: example }

tags:
  - name: Status
    description: Querying the configuration without revealing secrets
  - name: Forwarding
    description: Forwarding requests to the model providers

paths:
  /stato:
    get:
      tags: [Status]
      summary: Configured providers
      description: |
        Lists the providers for which the proxy holds at least one key.
        It reveals no key. The application uses it to show only the models that
        actually work, instead of letting them fail one by one.
      responses:
        "200":
          description: List of active providers
          content:
            application/json:
              schema:
                type: object
                required: [fornitori]
                properties:
                  fornitori:
                    type: array
                    items:
                      type: string
                      enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
              examples:
                twoKeys:
                  summary: Two providers configured
                  value: { fornitori: [groq, gemini] }
                none:
                  summary: Worker published but with no secrets
                  value: { fornitori: [] }
        "403":
          $ref: "#/components/responses/OrigineNonAmnessa"

```

```

/{fornitore}/{percorso}:
  parameters:
    - name: fornitore
      in: path
      required: true
      description: Destination provider
      schema:
        type: string
        enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
    - name: percorso
      in: path
      required: true
      description: |
        Path of the endpoint at the provider, reassembled as it is.
        Examples: `openai/v1/chat/completions` for Groq,
        `vlbeta/models/gemini-3.6-flash:streamGenerateContent` for Gemini.
      schema: { type: string }
      example: openai/v1/chat/completions
  post:
    tags: [Forwarding]
    summary: Forward a request to the provider
    description: |
      The body is transmitted unchanged. The proxy adds the key in the way
      that provider requires (an `Authorization: Bearer` header, `x-api-key`,
      or a query parameter) and, if the call fails for exhausted quota,
      retries with the next key within the same request.
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            description: Body expected by the provider's API, transmitted unchanged
          examples:
            groq:
              summary: Streaming completion on Groq
              value:
                model: llama-3.3-70b-versatile
                stream: true
                messages:
                  - role: user
                    content: Which IR bands identify a carboxylic acid?
    responses:
      "200":
        description: |
          The provider's answer, forwarded without modification (typically
          `text/event-stream` when `stream: true`).
        content:
          application/json:
            schema: { type: object }
          text/event-stream:
            schema: { type: string }
      "400": { $ref: "#/components/responses/CorpoIlleggibile" }
      "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
      "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
      "429": { $ref: "#/components/responses/TroppeRichieste" }
      "503": { $ref: "#/components/responses/ChiaveAssente" }
  get:
    tags: [Forwarding]

```

```
summary: Forward a read request to the provider
description: |
  It serves, among other things, the resolution of the Gemini model, which
  queries `ListModels` with GET and could not pass without a key in the
  browser.
responses:
  "200":
    description: The provider's answer
    content:
      application/json:
        schema: { type: object }
  "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
  "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
  "429": { $ref: "#/components/responses/TroppeRichieste" }
  "503": { $ref: "#/components/responses/ChiaveAssente" }
```

```
components:
  schemas:
    Errore:
      type: object
      required: [errore]
      properties:
        errore:
          type: string
          description: Message in Italian, meant to be shown to the user
  responses:
    OrigineNonAmmessa:
      description: The request's origin is not in the `ORIGINI` list
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Origine non ammessa da questo proxy." }
    FornitoreSconosciuto:
      description: The first path segment does not match a known provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Fornitore sconosciuto: pippo" }
    CorpoIlleggibile:
      description: The request body could not be read
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Corpo della richiesta illeggibile." }
    TroppeRichieste:
      description: The per-IP limit or the daily ceiling has been exceeded
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Troppe richieste da questo indirizzo: riprova fra un minuto." }
    ChiaveAssente:
      description: The proxy has no key configured for that provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example:
            errore: "Il proxy non ha una chiave per groq. Impostala con: wrangler secret put
            GROQ_KEYS"
  security: []
```

```
# CLIENT authentication is deliberately absent: the access control is the
# ORIGINI list plus the per-IP limits. Adding a client-side token would mean
# writing it into a static page, that is, making it public.
```

3. External APIs consumed by the application

All **optional**: with no network the application stays fully functional on local data, and says so to the user.

Service	Endpoints used	What it is for	If it does not answer
PubChem PUG-REST	<code>/compound/name/{name}/property/...</code> <code>
 /compound/smiles/{smiles}/SDF?record_type=3d</code> <code>
 /compound/cid/{cid}/PNG</code>	IUPAC name, formula, CAS, GHS pictograms, 3D conformers, 2D images	The app uses the local tabulated data and says so
RCSB PDB	<code>https://files.rcsb.org/download/{pdb}.pdb</code>	Protein structures for the 3D viewer	Explicit message in the viewer
NASA / ESA	Public images and data	Astrochemistry module	The module shows the tabulated data
AI providers	At the user's choice, or via the proxy	Assistant answers	The assistant declares it and offers a reachable alternative

None of these calls transmits personal data. There are no persistent identifiers and no active telemetry (`BSI_TELEMETRY_URL` is empty).

This is not asserted from the architecture alone: `audit_rete` seeds a canary value into the user's own data, uses the application across 6 pages and 87 sections, and inspects the URL, headers and body of every outgoing request.

4. Internal JavaScript APIs

Extension points for whoever has to work on the code. They are globals on `window` because the application is a set of classic scripts, not ES modules.

window.BSISpettri — spectral engine

Function	Signature	Description
gruppi	(smiles, callback)	Recognises functional groups via SMARTS/RDKit. Calls back with <code>null</code> if RDKit is unavailable: the caller decides whether to fall back or declare that it cannot compute.
gruppiSync	(smiles) → object null	Synchronous version: answers only if the molecule is already cached
bandeDaGruppi	(groups) → band[]	Table of IR bands {n, pk, w, i, f}
svgIR	(bands, options) → string	IR spectrum in transmittance, SVG, deterministic
legendaIR	(bands) → string	Assignments in HTML
quandoPronto	(callback)	Notifies when the analysis of a molecule is available, so it can be redrawn
precarica	()	Warms up RDKit while nobody is waiting

window.bsiProxy — proxy state

Function	Description
url()	Configured address, empty string if absent
stato()	Promise: queries <code>/stato</code> and records which providers are covered
copre(providerId)	Whether that provider is covered by the proxy
ricarica()	Re-reads the configuration

Other notable points

Symbol	Description
window.bsiLoadRDKit(cb)	Loads RDKit once only and calls back with the instance
window.irBandList(smiles)	IR bands — replaced by the new engine, keeps the original signature
window.makeIRsvg(bands, colour)	IR drawing — likewise
window.nmrPeakList(smiles)	¹ H peaks with integration
window.bsiCancellaDati(groups)	Selective deletion of the saved data
window.bsiOpenUpdate()	Opens the updates window
window.bsiAccentoLeggibile(colour, background, threshold)	Adjusts an accent so it reaches the WCAG threshold on that surface, preserving the hue
window.bsiEtichettaLeggibile(background, threshold)	The dual: picks the text colour from the background and, if neither extreme suffices, darkens the background while preserving the hue

Document updated to version bsi-v181.

Data Model — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v181
Purpose	Document where the data live, with what structure, and what lifecycle they have.

0. A necessary premise

BioSpecInfo has no server database. There is no backend, so there are no centrally managed relational tables, no migrations, and no users stored anywhere.

There is, however, a real data model, and an assessor is entitled to see it documented. It is organised across three levels:

Level	Technology	What it holds	Where it lives
Preferences and progress	localStorage	Settings, API keys, AI history, study progress	The user's browser
Document archive	IndexedDB	Photos, documents, HTML sections, notes, activity log	The user's browser
Teaching datasets	SQLite via sql.js	Exercise tables, created in memory at each session	Memory, not persisted

All data stay on the device. They are not transmitted, there is no server-side user identifier, and the user can delete them selectively from the interface.

1. Entity diagram

```
erDiagram
    DISPOSITIVO ||--|| PREFERENZE : "possiede"
    DISPOSITIVO ||--|| ARCHIVIO : "possiede"
    DISPOSITIVO ||--o{ SESSIONE_SQL : "apre"

    PREFERENZE {
        string bsi_ai_provider "fornitore AI scelto"
        json bsi_api_keys "chiavi per fornitore"
        string bsi_proxy_url "indirizzo del proxy"
        json bsi_ai_threads "conversazioni salvate"
        json bsi_ai_memory "memoria dell-agente"
        json bsi_prov_ko "fornitori irraggiungibili"
        json bsi_modelli_ko "modelli ritirati"
        json bsi_tetti "tetti di token appresi"
        json bsi_srs "ripetizione spaziata"
        json bsi_quiz_progress "avanzamento quiz"
        string bsi_section "ultima sezione aperta"
    }

    ARCHIVIO ||--o{ PHOTOS : contiene
    ARCHIVIO ||--o{ DOCS : contiene
    ARCHIVIO ||--o{ SECTIONS : contiene
    ARCHIVIO ||--o{ NOTES : contiene
    ARCHIVIO ||--o{ ACTIVITY : registra

    PHOTOS {
        int id PK "autoIncrement"
        string name
        string type "MIME"
        int size "byte"
        string data "data URL"
        int added "epoch ms"
    }
    DOCS {
        int id PK "autoIncrement"
        string name
        string type
        int size
        string data "data URL"
        int date "epoch ms"
    }
    SECTIONS {
        int id PK "autoIncrement"
        string name
        string icon
        string html "contenuto"
        int size "caratteri"
        int added "epoch ms"
    }
    NOTES {
        string key PK "sempre main"
        string text
        int updated "epoch ms"
    }
    ACTIVITY {
        int id PK "autoIncrement"
        string icon
```

```

string msg
string time "data locale"
}

```

The diagram keeps the entity and field names as they appear in the code, in Italian: renaming them in the translation would document a model that does not exist.

2. Preferences and state — localStorage

2.1 AI assistant

Key	Type	Content	Lifecycle
bsi_ai_provider	string	Identifier of the selected provider	Until the user changes it
bsi_api_keys	JSON	{ providerId: key }	Until explicitly deleted
bsi_api_key	string	Single key, legacy format	Migrated to bsi_api_keys
bsi_proxy_url	string	Proxy address	Until "Disconnect"
bsi_ai_threads	JSON	{ threads: [...], activeId } — max 30 chats, 100 messages each	Pruned automatically
bsi_ai_memory	JSON	Facts the agent remembers between sessions	Deletable
bsi_ai_history	JSON	Condensed history	Deletable
bsi_nucleo	string	"Core" mode active	—

2.2 What the assistant learns about providers

These three keys are the reason the assistant improves with use instead of repeating the same mistakes. They have an expiry because today's failure must not be a permanent sentence.

Key	Structure	Expiry	What it is for
bsi_prov_ko	{ providerId: { t: epoch, n: count } }	24 hours	Providers that do not answer from this device. Demoted in the selection, never removed.
bsi_modelli_ko	{ digest: { modelli: [...], t } }	7 days	Models retired by the provider, so as not to offer them again. Indexed on the digest of the key.
bsi_tetti	{ providerId: tokenCeiling }	Permanent	Token ceiling learned from the error message rather than hard-coded

2.3 Study and progress

Key	Content
bsi_srs	Spaced repetition: cards, intervals, next review

Key	Content
<code>bsi_sm2</code>	Parameters of the SM-2 algorithm
<code>bsi_quiz_progress</code> , <code>bsi_quiz_history</code>	Quiz progress and history
<code>bsi_studypath</code>	The study path chosen
<code>bsi_notes</code> , <code>bsi_note_<section></code>	Personal notes, global and per section
<code>bsi_ds_prog</code> , <code>bsi_ds_proj</code>	Progress in the data-science module
<code>bsi_section</code>	Last section opened, to resume from there
<code>bsi_guide_v3</code>	Introductory guide already seen

2.4 Identifiers and licence

Key	Content	Privacy note
<code>bsi_device_id</code>	Random local identifier	Not transmitted. It only distinguishes installations within the same browser.
<code>bsi_user_email</code>	Email, if entered by the user	Entered voluntarily, not transmitted
<code>bsi_pro_license</code> , <code>bsi_trial_start</code>	State of Pro mode	Local
<code>bsi_rdkit_lib2</code>	Molecule library of the laboratory	Local

That none of these ever leaves the device is verified directly by `audit_rete` , which seeds a canary value into these very keys and then inspects every outgoing request.

3. Document archive — IndexedDB

Database name	<code>bsi_filemanager_v1</code>
Schema version	<code>2</code>
Object stores	<code>photos</code> , <code>docs</code> , <code>sections</code> , <code>notes</code> , <code>activity</code>

3.1 Structure of the stores

Store	Key	Fields	Notes
<code>photos</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>type</code> , <code>size</code> , <code>data</code> , <code>added</code>	Images as data URLs. Per-file limit: 20 MB
<code>docs</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>type</code> , <code>size</code> , <code>data</code> , <code>date</code>	Generic documents
<code>sections</code>	<code>id</code> (autoIncrement)	<code>name</code> , <code>icon</code> , <code>html</code> , <code>size</code> , <code>added</code>	HTML pages uploaded by the user

Store	Key	Fields	Notes
notes	key	key ("main"), text , updated	A single entry
activity	id (autoIncrement)	icon , msg , time	Log of the most recent operations

The `files` store, present in earlier schema versions, is no longer created from version 2.

3.2 Opening and unavailability

Opening IndexedDB is **asynchronous**, and in some configurations (Safari in private browsing, site data blocked) it **never happens**. The code awaits a promise rather than reading a variable that might not yet be set, and distinguishes two cases:

- **reading** with the store absent → returns an empty set, the page shows

"nothing yet";

- **writing** with the store absent → the error reaches whoever pressed the

button, and a persistent warning appears at the top of the page saying nothing will be kept.

If another tab requests a version upgrade, the connection is closed (`onversionchange`) so as not to block both.

4. Teaching datasets — SQLite in memory

The data-exercise module creates an SQLite database **in memory** through `sql.js` (SQLite compiled to WebAssembly). The tables are generated at each session from the datasets bundled in the repository and are **not persisted**: closing the tab makes them vanish.

The choice is deliberate: the exercise must start from a known state every time, and no user data end up in a database they did not create.

5. Reference data — constants in the code

The scientific data are not in a database but in constant structures inside `index.html` , loaded with the application. They are read-only and versioned together with the code.

Structure	Content	Verification
FARM_DATA	233 drugs : name, category, SMILES, molecular weight, mechanism of action, indications, adverse effects, class	<code>tools/verifica-farmaci.js</code> — structure ↔ molecular weight
UV_DATA	Chromophores, λ_{max} , extinction coefficients, Woodward's rules	<code>audit_dati</code>
MS_DATA	Characteristic fragmentations by class	<code>audit_dati</code>
MOLECOLE_ESAME	Tabulated exam molecules	<code>test_spettri</code>

Structure	Content	Verification
PHYS_SIMBOLI	Physical constants with symbol and unit	test_costanti
SMARTS , SMARTS_H	Patterns for functional-group recognition	test_spettri

6. Lifecycle and deletion

The user can delete data **by group**, from the interface, without wiping everything. The groups are defined in `bsiCancellaDati()`.

Group	What it removes
chiavi	API keys, chosen provider, proxy address, memory of unreachable providers
chat	Conversations and the assistant's memory
studio	Spaced repetition, quizzes, study path, notes
tutto	Every <code>bsi_*</code> key and the entire IndexedDB archive

Deleting the archive proceeds even if a single entry fails, and reports how many were not removed: a "delete everything" operation that stops halfway without saying so would be worse than the problem it solves.

7. Retention and space limits

Constraint	Value	Behaviour when exceeded
localStorage	5–10 MB depending on the browser	Writes fail. The app stays operational: chats are pruned (30 chats, 100 messages), and every write is guarded.
IndexedDB	Quota granted by the browser	Error reported to whoever pressed, with a persistent warning
Single image	20 MB	Refused with an explicit message; the upload of the others continues

Behaviour with space exhausted is verified by the `audit_quota` and `test_filemanager` benches: ten pages out of ten stay usable with `localStorage.setItem` forced to fail.

Document updated to version `bsi-v181`.

Deploy Guide — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v181
Purpose	Operational procedure for publishing, updating and rolling back the application.

0. What gets published, in practice

BioSpecInfo is a set of **static files**. There is nothing to compile, no process to keep running, no database to provision.

Build	None. The repository files <i>are</i> the application.
Server runtime	None
Environment variables	None for the application. Only the optional proxy has any (\$4).
Database	None server-side
Server requirement	Serve static files over HTTPS with the correct MIME types

The one non-obvious requirement: the server must return `application/wasm` for `.wasm` files, otherwise RDKit and SQLite do not start. GitHub Pages, Netlify, Vercel and Cloudflare Pages already do.

1. Environments

Environment	How it is obtained	What it is for
Development	<code>python3 -m http.server 8899</code> from the root	Daily work and verification benches
Preview	A branch published on Pages, or <code>npx serve</code>	Checking before merging
Production	GitHub Pages from the <code>main</code> branch	<code>samupropiol-ship-it.github.io/BioSpecInfo-v11/</code>

Why a server is needed even in development. Opening `index.html` with a double click (`file://`) does not work: RDKit and SQLite are WebAssembly modules and require an HTTP context. It is the reason the verification benches also run against a local server.

2. Publishing to production

2.1 Procedure

```
# 1. from the working branch, the full battery
python3 -m http.server 8899 &
node tools/genera-evidenza.js          # must close with 0 failures

# 2. bump the version – TWO lines, always as a pair
#   sw.js:      var CACHE = 'bsi-v<NNN+1>';
#   index.html: window.BSI_APP_VERSION='bsi-v<NNN+1>';

# 3. regenerate SBOM, PDFs, packages and evidence on the new version
node tools/genera-sbom.js
node tools/genera-pdf.js               # 36 PDFs; every footer carries version and commit
node tools/genera-pacchetti.js         # the three delivery packages; fails if a link is broken
node tools/genera-evidenza.js

# 4. merge into main
git add -A && git commit -m "..."
git push -u origin <branch>
#   then pull request and squash merge
```

GitHub Pages publishes automatically from `main`: the *pages build and deployment* workflow takes about a minute.

2.2 The two version lines

`CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` **must match**.

- `CACHE` determines whether the Service Worker considers the cache stale:

without a bump, devices keep serving the old files and the publication reaches nobody.

- `BSI_APP_VERSION` is what the user sees in the "Updates" window and what the update check compares with the `sw.js` on GitHub.

They drifted apart once in the past — the second had been stuck for months — and the result was that the update check was comparing the wrong number.

2.3 How the update reaches users

1. The browser detects a different `sw.js` and installs the new Service Worker.
2. `skipWaiting()` activates it at once; the old caches are deleted.
3. The Service Worker **notifies** the open tabs (`BSI_SW_UPDATED`) — it does not reload them on its own authority.

1. The page decides: if no work is in progress it reloads itself; if the assistant is answering or there is unsaved text, it shows "Update now / Later".

Whoever opens the app at that moment already receives the new version.

3. Publishing alternatives

The application does not depend on GitHub Pages. Any static file host will do, provided it is over HTTPS (the Service Worker requires it).

Platform	Command / configuration	Notes
GitHub Pages	Settings → Pages → branch <code>main</code> , folder <code>/</code>	In use. Free, custom domain possible
Netlify	<code>netlify deploy --prod --dir=.</code>	No build command
Vercel	<code>vercel --prod</code>	Static project, no framework
Cloudflare Pages	<code>npx wrangler pages deploy .</code>	Convenient if the proxy is already in use
VPS with nginx	Copy the files to <code>/var/www/biospecinfo</code>	See the configuration below

nginx configuration

```
server {
    listen 443 ssl http2;
    server_name biospecinfo.example.org;
    root /var/www/biospecinfo;

    # Without this MIME type, RDKit and SQLite do not start
    types { application/wasm wasm; }

    # sw.js must not be cached by the CDN: it is what announces new versions
    location = /sw.js {
        add_header Cache-Control "no-cache, no-store, must-revalidate";
    }

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

A rule that holds everywhere: `sw.js` must not be cached

If a CDN keeps `sw.js` , the browser keeps seeing the old version and **no update ever arrives**, however much one publishes. This holds on every platform.

4. Optional proxy — a separate deployment

The proxy is a **distinct** component, owned by the user, published on their own Cloudflare account. It is not required in order to use the application.

```
cd proxy
npx wrangler login
npx wrangler deploy
# the address is printed: https://spectra-proxy.<account>.workers.dev

npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
npx wrangler secret put GEMINI_KEYS   # optional
```

The address is then pasted into the assistant's "Proxy" box, inside the app. The same four steps are available **inside the application**, with the commands ready to copy.

Recommended configuration

```
# Without this variable the proxy accepts requests from any origin
npx wrangler secret put ORIGINI
# value: https://samupropio1-ship-it.github.io
```

Variable	Default	Why change it
ORIGINI	(empty = all)	Always set it. An open proxy is usable by anyone, at the expense of your quota.
LIMITE_IP	20	Requests per IP within the window
TETTO_GIORNO	2000	Overall daily ceiling

Keys must never go into `wrangler.toml`. That file is versioned: a key written there ends up on GitHub, and the bots that scan it find it within hours. Use `wrangler secret put`, which keeps them out of the repository.

5. Domain and certificate

Aspect	Situation
HTTPS	Mandatory: without it the Service Worker does not register and the app does not work offline
Certificate	Provided and renewed automatically by GitHub Pages
Custom domain	Settings → Pages → Custom domain, plus a <code>CNAME</code> record pointing to <code><user>.github.io</code>

6. Monitoring

There is no server to monitor: there are no processes that could fall over, nor resources that could run out.

What to watch	How
Outcome of the publication	GitHub → Actions → <i>pages build and deployment</i>
Availability	An HTTP check on <code>index.html</code> (UptimeRobot or similar)
Client-side errors	The browser console. Remote telemetry exists but is disabled (<code>BSI_TELEMETRY_URL</code> empty): enabling it requires a deliberate decision about data handling.
Proxy consumption	Cloudflare → Workers → Analytics

7. Rollback and emergency procedures

7.1 Going back to the previous version

```
git revert <commit>
git push origin main
```

Pages republishes in about a minute. Devices receive the restored version as a normal update — **provided `CACHE` changes**: a revert that also takes the version number backwards is fine, because what counts is that the value be *different*, not that it be greater.

7.2 "I published but I do not see the changes"

In order:

1. **Was the version bumped?** It is the cause in ninety per cent of cases.

Without it, the Service Worker serves the cache.

1. **Has Pages finished?** GitHub → Actions.
2. **Is `sw.js` cached by a CDN?** It must answer `no-cache`.
3. From the app: ✨ menu → **Updates** → *Check*, which forces a `registration.update()`.

1. As a last resort: developer tools → Application → Service Workers → *Unregister*, then reload.

7.3 "The assistant has stopped answering"

This is not a publishing problem: it concerns the AI provider. In the app, the 🦊 **Test** button measures in a few seconds which providers the device can actually reach, without needing any key.

7.4 Restoring the user's data

The data live on the device and **there are no copies on a server**: they cannot be restored remotely. This is a direct consequence of the local-first architecture, and it is declared to the user. Exports (notes, documents) remain the only form of backup.

8. Pre-publication checklist

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` bumped and **equal**
- ☐ `node tools/genera-sbom.js` re-run
- ☐ `node tools/genera-pdf.js` re-run — otherwise the attachments describe the previous version
- ☐ `node tools/genera-pacchetti.js` re-run and closed with **0 broken links**
- ☐ Evidence report regenerated against the commit being published
- ☐ Any new deviations noted in `09-Release-Conformance-Statement.md` §4
- ☐ No conflict markers (`verifica_guida` checks this across 58 files)
- ☐ No API key in the versioned files
- ☐ The accessibility debt **has not grown** — if it has fallen, regenerate the

baseline with `node tools/verifica-accessibilita.js --aggiorna-riferimento` and record it in the commit

- ☐ Code coverage **has not fallen** — `audit_copertura` fails if it has

Document updated to version `bsi-v181`.

Functional Specifications — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v181
Purpose	Describe what the product does, for whom, under which rules and with which limits.

1. Product description

BioSpecInfo is an **interactive cheminformatics** platform for university study of chemistry, biochemistry, pharmacology and astrochemistry.

The whole scientific logic — interpretation of molecular structures, spectral prediction, calculations, 2D and 3D visualisation — runs **in the student's browser**. There is no server: the application installs as an app, works with no network, and transmits no data.

1.1 Objectives

Objective	How it is pursued
Total accessibility	No registration, no cost, no server. Whoever opens the link is already using it.
Independence from the network	Everything needed is bundled and cached: it works in a lecture hall, on a train, in a library with no signal.
Verifiable correctness	Scientific data are put through automatic checks; the limits of the models are declared rather than hidden.
Privacy by construction	There is no backend that could collect data: the absence of collection is not a promise, it is a consequence of the architecture.

1.2 What the product is not

Not a medical device	It does not support diagnosis, prescription or dosing (Reg. EU 2017/745)
Not a certified analytical instrument	The spectra are predictions, not measurements
Not a collaborative platform	No accounts, no sharing, no synchronisation between devices

2. Actors

Actor	Description	What they can do
Student	Primary user. Studying for a university exam.	Every function: consultation, calculation, quizzes, spaced repetition, AI assistant

Actor	Description	What they can do
Teacher	Uses the app as teaching support	The same functions; uploading their own material into the File Manager
Technical assessor	A company or committee examining the work	Reading the documentation and re-running the verification benches

There are no technical roles: there is no authentication, so there is no distinction of permissions. The File Manager has a local protection that is a **deterrent**, not an access control (see §6).

3. Functional areas

The application has **89 sections**, grouped by area of study.

3.1 Spectroscopy

Function	Description
Spectroscopy centre	A molecule is entered (SMILES, name, formula) and one obtains IR, $^1\text{H}/^{13}\text{C}$ NMR, MS, UV-Vis, Raman, 2D and 3D structure, functional groups
Visual IR spectra	Characteristic bands by class of compound
Mass spectrum	Fragmentation patterns and diagnostic rules
UV-Vis	Chromophores, λ_{max} , Woodward's rules
NOE and 2D NMR	Spatial correlations
Reference tables	Tabulated values for quick consultation
Spectrum quiz	Recognising the functional group from a spectrum

3.2 General and physical chemistry

Interactive periodic table · chemical calculator · reaction balancing · degrees of unsaturation · distribution diagrams · VSEPR · molecular orbitals · advanced thermodynamics · electrochemistry (Nernst) · pKa and buffers · laboratory calculator · constants and formulae.

3.2-bis Separation — chromatography

Chromatography section: a chromatogram computed from the parameters (theoretical plates, retention factor, selectivity, dead time) with an overview and an expanded view of the peak pair; a **van Deemter** curve with the three terms separated and the minimum computed in closed form; a **Kováts retention index** calculator; a table of the quantities with their working ranges; GC/HPLC comparison; a guide to detectors; diagnostics for peak-shape defects.

Rule applied. The chromatogram is not an illustration: the Gaussians have $\sigma = t_{R}/\sqrt{N}$ and the resolution shown is the one that would be measured on those peaks. The section explicitly declares what it does **not** do — predict a molecule's retention time from its structure, which would require experimental parameters of the stationary phase that cannot be derived from the SMILES alone.

3.2-ter Calibration and uncertainty

Calibration section: ordinary least-squares regression on data pasted from a spreadsheet, with slope and intercept **accompanied by their standard deviation**; a plot of the line with a 95 % confidence band and a **residual plot**; the unknown's concentration with a full confidence interval (the three sources $1/M$, $1/N$ and distance from the centroid are shown separately); **LOD and LOQ** per ICH Q2(R1); uncertainty propagation indicating which contribution dominates; significant-figure rules.

Two diagnostics that an r^2 does not provide:

Diagnostic	Method	Why r^2 is not enough
Non-linearity	F-test between line and parabola on the same data	An r^2 of 0.984 accompanies an obvious saturation curve: r^2 measures how much variance the line explains, not whether the line is the right model
Outlier	Deleted residual: the judging line is rebuilt without the point being judged	The naive criterion `

The second row is a trap met while writing the section, not a textbook case: the 3σ criterion was already written, and the example meant to demonstrate it was not being flagged. The check passed without seeing anything.

3.2-quater Cheminformatics and QSAR

Cheminformatics section: a workbench that goes from the raw set to a verdict on the model, entirely in the browser. SMILES are pasted in — with a name and an activity, where there are any — and six panels return what a cheminformatics group produces in a day.

Panel	What it does
Descriptors	Standardisation (largest fragment, canonical SMILES, deduplication), 43 RDKit descriptors, Lipinski's rule of five, Veber filter, QED recomputed — MinimalLib does not expose it
Similarity and groups	Morgan/ECFP4, RDKit, pattern and MACCS fingerprints; Tanimoto and Dice; similarity matrix; nearest neighbours; Butina clustering and MaxMin selection for diversity
Chemical space	PCA on the standardised descriptors, with explained variance per component and the loadings that say which descriptor moves which axis; Bemis-Murcko scaffolds with the molecule count per scaffold

Panel	What it does
QSAR model	Kernel ridge (Tanimoto kernel), kNN over Tanimoto or logistic regression; scaffold split ; grouped cross-validation over 3, 5 or 10 folds; model comparison on identical folds with the trivial reference at the top; hyperparameters chosen by nested validation ; conformal prediction intervals ; learning curve ; applicability domain
Search	Substructure search across the whole set with RDKit's <code>SubstructLibrary</code> , SMARTS or SMILES query, occurrence counts, eight ready queries
SAR table	R-group decomposition around a core: one row per molecule, one column per substitution position, with each substituent's median activity. Molecules without the core are excluded , not shown with empty cells
Matched pairs	Pairs differing by a single substitution, grouped by transformation with median, range and number of concordant pairs
Activity cliffs	Structurally close pairs with distant activities, ranked by SALI : these are the pairs every model gets wrong, and it is honest to show them
Structural alerts	PAINS and Brenk, with the matched fragment highlighted and the reason it is flagged

Three choices that separate a workbench from a demonstration:

Choice	Method	Why
Scaffold split	The test set contains only scaffolds never seen in training	A random split puts close analogues on both sides: the model recognises rather than predicts, and the R^2 that comes out does not survive the first new molecule
Null model by scrambling	The same model is retrained on shuffled labels , eight times; the verdict is the comparison	An R^2 of 0.4 can arise by chance when the molecules are few. If the random twins reach the same score, the model has learnt nothing — and the panel says so , instead of showing only the good number
Deduplication on the canonical form	<code>OC(=O)C</code> and <code>CC(=O)O</code> are the same molecule	The same molecule in training and in test is an information leak, and the written string is not enough to see it

The verdict is written out in words, not left to the reader. The QSAR panel does not stop at the metrics: it states whether the model beat all its random twins, and by how much. Measured on the 28-inhibitor example: R^2 0.861 against a maximum of 0.574 among the eight twins — margin 0.287. On the same molecules with random labels the model does **not** beat the twins, and the panel says so in the same way.

The results can be taken away

Two exports, both formed in memory by the browser: there is no network request, which is why the note beside the buttons can say that no data leaves the device without asking anyone to take it on trust.

Export	Content
CSV table	One row per molecule: name, canonical and original SMILES, activity, scaffold, nine descriptors, QED, Lipinski violations, Veber outcome, alerts found. With cross-validation active, predicted and residual are added — and only then, because cross-validation is the one scheme that gives every molecule a prediction from a model that had not seen it
SDF	The structures with generated 2D coordinates , the name on each block's first line and the data fields attached: it is the format in which sets are exchanged between groups and programs. Verified by round-trip
Method report (.md)	What is needed to redo the same analysis: RDKit version, fingerprint, split type, number of molecules and scaffolds, the seed , the metrics, the null-model verdict, the folds one by one, and the declared limits

Why the seed is in the report. Without it, "random split" and "scrambled labels" mean nothing: two runs would give different numbers and nobody could say which was the right one. The generator is a seeded mulberry32, and the same seed gives the same folds.

A measurement no other bench could make

`verifica-accessibilita` walks the 89 sections and measures what is **visible**. The six panels of this section live inside a container that stays `display:none` until the analysis has been run: that bench never saw them, and its "contrast 0" was not speaking about them.

`test_cheminfo` runs the analysis for real, opens the six panels in turn and measures WCAG contrast on every piece of text that appears there — **460 elements, 0 defects**. The formulas are rewritten inside the bench rather than taken from the application: a bench that asked the code under test how good its own contrast was would be measuring nothing.

The engine lives in `bsi-cheminfo.js`, is exposed as `window.BSIChem` and is verified by the `test_cheminfo` bench (189 checks) against hand-computable values.

3.3 Organic chemistry

Retrosynthesis (98 exercises) · synthesis and mechanisms with electron arrows · animated mechanisms · structure editor · *Organic Chemistry 1+2* module with 25 modules.

3.4 Biochemistry

Amino acids · biomolecules · animated metabolic pathways · enzyme kinetics (Michaelis-Menten) · 3D macromolecules · interactive biological structures · biochemistry guide · exam biochemistry.

The **biochemistry guide** is a document of its own (`Biochimica_Guida_Definitiva.html`, 8.2 MB) loaded into a frame by the section: 12 chapters plus the index, 78 subsections, 98 embedded figures, and two chapters devoted to the exam — "Oral exam guide" and "Real exam questions and correct answers". It contains **no scripts and makes no network requests**: it cannot raise JavaScript errors, and the alternative text of the 98 figures comes from the captions the author wrote, not from invented descriptions (see `docs/06 §3-ter`).

The document closes with a **self-test of 12 multiple-choice questions**, recovered from the previous guide — where it was a JavaScript quiz — and rewritten **without a line of code**: the four options are in plain sight, the solution sits in a native `<details>` that opens on a click. In print the `<details>` disappears and a dedicated copy of the solution appears, so the printed copy does not lose the explanations.

3.5 Pharmacology

Function	Description
3D Atlas	One chooses a body region, a disease and a drug: the model highlights the target organ
Database	233 drugs with structure, molecular weight, mechanism, indications, adverse effects, class
Interactions	Checking interactions between active ingredients
Pharmacokinetics	Calculator for half-life, clearance, volume of distribution
Clinical cases	Diagnosis-therapy paths for teaching purposes

3.6 Astrochemistry

Molecules of the interstellar medium · exoplanets and JWST data · nebulae · stellar spectra · nucleosynthesis · cometary chemistry · quizzes.

3.7 Study tools

Function	Description
Spaced repetition	SM-2 algorithm on cards generated from the contents
Quizzes	By subject, with score and history
Study path	A guided sequence of topics
Notes	Global and per section
Pomodoro	Study timer
File Manager	Personal archive of photos, documents and HTML pages

3.8 The "Spectra" assistant

An AI agent with **35 tools** that can interrogate the application itself: compute descriptors, generate spectra, look up constants, balance reactions, open sections. It supports ten providers, and stays usable when one of them fails (§5.2).

4. Main flows

4.1 Analysing a molecule

1. Open the Spectroscopy centre
2. Enter the molecule:
 - SMILES → used directly
 - name or formula → resolution via PubChem (needs network)
 - drawing in the editor → SMILES generated
3. The application shows: 2D structure, functional groups, descriptors, predicted spectra (IR, NMR, MS, UV-Vis), drug-likeness
4. The 3D structure loads if there is a network (conformer from PubChem)

With no network: point 3 stays complete for SMILES entered directly; resolution by name and the 3D structure are unavailable, and the application says so instead of hanging.

4.2 Asking the assistant

1. Tap the "Spectra" button
2. On first use: choose a provider and enter an API key, or connect a proxy (and then no key is needed)
3. Type the question
4. The assistant answers, using its tools when needed: every tool used is shown, not hidden

If the provider does not answer: the assistant notes it, demotes it in the selection for 24 hours and moves to another configured provider. The 🦋 **Test** button measures in a few seconds which providers the device actually reaches, **without needing any key**.

4.3 Studying with spaced repetition

1. Open the study section
2. Generate a deck from the contents
3. For each card, rate how well it was remembered
4. The SM-2 algorithm computes when to show it again
5. On reopening, only the cards that are due appear

4.4 Archiving your own material

1. File Manager → local password
2. Upload photos, documents or HTML pages
3. The material stays ON THE DEVICE (IndexedDB), it is not transmitted

5. Business rules

5.1 Scientific data

Rule	Rationale
A molecular structure enters the database only if the recomputed weight agrees with the literature one (± 0.6 u)	A plausible but wrong datum is indistinguishable from a correct one for the user
If a structure does not pass verification, the entry stays without a structure	A wrong datum replaced by another wrong one is no progress
Every absence must be recorded with its reason	An unjustified exception becomes a shortcut
Spectra are deterministic	Two spectra of the same molecule must be comparable
An ambiguous physical constant is not resolved at random	A wrong value given with confidence is worse than a request for clarification

5.2 AI assistant

Rule	Behaviour
A provider that does not answer is remembered for 24 hours	Then it is retried: today's failure is not a sentence
A retired model is learned from the error message	Hand-coded lists grow stale
An imposed wait longer than 10 seconds switches provider, if a ready one exists	Beyond that threshold the user believes the app has frozen
One turn at a time	Two overlapping submissions corrupt the history
Paid providers are not offered as free ones	Without active billing one hits the limit at the first question

5.3 User data

Rule	Behaviour
At most 30 conversations, 100 messages each	Beyond that, <code>localStorage</code> fills up and every subsequent save fails
Images up to 20 MB	Beyond that, explicit refusal; the others proceed
Selective deletion by group	Wiping everything to free space is an avoidable loss
A failed write must be declared	"  Saved" when it is not true is worse than an error

6. Current limitations

Declared, not hidden. The detail is in [09-Release-Conformance-Statement.md](#) §4.

#	Limitation	Practical consequence
1	2 entries with no structure to show (it was 6)	Ivermectin is a mixture of B1a/B1b homologues and Coartem a combination of two active ingredients: a single SMILES notation does not represent them, and for these two the 2D/3D and predicted spectra are missing. Digoxin, vincristine and tacrolimus (topical and systemic) now have their structure, taken from ChEMBL and verified against the molecular weight
2	Predicted, not measured, spectra	Useful for recognising functional groups; not for formal identification
3	No synchronisation	Data do not pass from one device to another, and there are no copies on a server
4	File Manager: deterrent, not security	On a static site, whoever reads the source bypasses any page-side control. The source carries only the SHA-256 digest of the password, never the password
5	Functions that need a network	IUPAC name, CAS, GHS, 3D conformers, AI assistant
6	Accessibility automated, not complete	Contrast, accessible names, labels, heading hierarchy across 13 pages and all 89 sections: 0 defects , including text over gradients (judged against the worst stop). What stays outside is text inside SVGs and text over a real background image, counted on every run, and everything requiring human judgement
7	Chromatography: no retention prediction	The section computes resolution, efficiency and indices given k , α and N ; it does not predict k from a structure, which would require experimental parameters of the stationary phase
8	Partial code coverage	Measured: 49.89 % of statements over the widest path a bench walks. It is not whole-battery coverage, and it is statements, not branches; see 08-Traceability-Matrix.md §8
9	Verification on Chromium only	Firefox and WebKit are tested by hand, not by a bench

7. Possible developments

Not commitments: directions consistent with the architecture.

Area	Proposal	Why
Data	The 2 remaining entries have no structure to show	It is not a gap to close: ivermectin is a mixture, Coartem a combination. At most one could show the components , declared as such
Spectra	Comparison with experimental databases (SDBS/NIST) inside the app	It would make the distance between prediction and measurement visible
Accessibility	Add <code>axe-core</code> alongside, for the rules the bench does not implement	ARIA roles, tab order and focus management stay outside the current measurement
Export	Full export of the user's data	The only possible remedy for the absence of backups

Area	Proposal	Why
Assistant	A preconfigured proxy for those who do not want to publish one	It would remove the last barrier to AI access

Document updated to version bsi-v181.

A guide for the people who use it — BioSpecInfo

This guide is written for people who study, not for people who program. You do not need to know anything about computing.

What it is

BioSpecInfo is an application for studying chemistry, biochemistry, pharmacology and astrochemistry. It opens like a website, but you can **install** it on your phone or computer and it works **without internet** too.

Three things to know straight away:

- **No sign-up needed.** No account, no email, no password.
- **It costs nothing.**
- **Your data stay yours.** Notes, files and conversations never leave your

device: there is no server they could reach.

👉 samupropio1-ship-it.github.io/BioSpecInfo-v11

Installing it (optional, but worth it)

Installing gives you an icon on your screen, full-screen opening and offline operation.

Device	How to do it
Android / Chrome	: menu → <i>Install app</i> (or <i>Add to Home screen</i>)
iPhone / iPad	Share button → <i>Add to Home Screen</i>
Computer	Install icon in the address bar, on the right

After installing, open the app once with the network on: it will download everything it needs. From then on it works with no signal too.

Finding your way around

Element	Where it is	What it is for
 Search	At the top	Searches the whole app. Shortcut: <code>Ctrl+K</code>
Section menu	Navigation bar	The 89 sections, grouped by subject
	Bottom right	Tools, settings and updates
Spectra	Bottom right	The assistant that answers questions

The things you will do most often

Analysing a molecule

1. Open the **Spectroscopy centre**
2. Type the molecule into the box. You can use:
 - the **name** — aspirin, caffeine
 - the **SMILES formula** — CC(=O)Oc1ccccc1C(=O)O
 - or **draw it** with the editor
1. Press Enter

You get the 2D and 3D structure, the functional groups, the molecular weight, the predicted spectra (IR, NMR, mass, UV-Vis) and the pharmacological properties.

If you have no network: searching by *name* does not work, because it has to consult an external database. The SMILES formula, on the other hand, always works.

Reading an IR spectrum

The spectrum is drawn as on a real instrument: the scale at the bottom runs from 4000 to 400 cm^{-1} (**from left to right the numbers go down**) and the bands point **downwards**.

The coloured area below 1500 cm^{-1} is the **fingerprint region**: that is where the specific molecule is recognised, not just its class. Below the chart you will find the list of bands with their assignment.

The spectra are **predicted** from the structure, not measured in a laboratory. They are there to help you recognise functional groups and to understand the link between structure and spectrum. For experimental data, the app sends you to the official databases.

Studying with cards

1. Open the study section
2. Generate a deck of cards
3. For each card, after answering, say how well you remembered it

The app works out **when** to show it to you again: the things you know well come back rarely, the uncertain ones come back soon. When you reopen, you only find the cards that are due.

Looking up a drug

Open **Pharmacology**. For each of the 233 drugs you will find the structure, molecular weight, mechanism of action, indications, adverse effects and class.

 **For teaching purposes only.** This information does not replace the judgement of a doctor or a pharmacist, and is not there to decide a therapy.

Spectra, the assistant

Spectra answers chemistry questions and can use the app's own tools: calculate, generate spectra, look up constants, balance reactions.

Getting it working

Spectra relies on an external artificial-intelligence service, and you need a free **key**. Two routes:

The simplest — a key of your own

1. Go to console.groq.com and create a key

(free, it takes two minutes)

1. Open Spectra, paste the key into the  box
2. Done

The definitive one — a proxy

If services often fail to answer for you, or you would rather not keep the key on your phone, you can publish a *proxy*: a small free program that keeps the key safe and makes even the services that are blocked from the browser work.

In the assistant's **Proxy** box you will find "**How to set it up, in 4 steps**", with the commands ready to copy.

When Spectra does not answer

Almost always it is not the app's fault but the external service's. Press  **Test** in Spectra's panel: in a few seconds it tells you **which services your device can actually reach** — and you need no key to find out.

If a service stops answering, Spectra remembers it for 24 hours and moves it to the bottom of the list, under "*Did not answer from this device*". It does not delete it: tomorrow it might work again.

The File Manager

A personal archive for photos, documents and HTML pages, protected by a password.

 **Something you need to know.** This protection is a **deterrent**, not real security: whoever knows where to look can get around it. Do not put confidential documents in it.

The files stay on your device. **There is no copy anywhere else:** if you clear your browser data or change phone, they are gone. Use the export for the things you care about.

Frequently asked questions

Do I have to pay for anything? No. The app is free. If you use Spectra, the AI service's free key has daily limits, but nothing is paid.

Does it really work without internet? Yes, after the first opening. The only things left out are those that by their nature have to ask someone else: searching molecules by name, 3D structures downloaded on the spot, and Spectra.

Are my notes saved somewhere? Only on your device. There is no server: nobody can read them, but nobody can recover them for you either if you lose them.

How do I delete my data? ✨ menu → Settings → delete data. You can choose **what**: only the keys, only the conversations, only the study data, or everything.

I updated but I do not see anything new. ✨ menu → **Updates** → *Check*. There you can also see which version you are running.

Can I trust the spectra for a report? For recognising functional groups and understanding the structure-spectrum relationship, yes. For a figure to be cited as experimental, no: they are predictions. The app sends you to the official databases (NIST, SDBS) with direct links.

I found an error in a chemical datum. Report it: open an *issue* on the [GitHub repository](#). The data go through automatic checks, but no check finds everything.

If something goes wrong

Problem	What to try
The app does not open	Reload it. If it persists, ✨ menu → Updates → Check
I do not see the new features	The version is in the ✨ menu → Updates
Spectra does not answer	The 🐛 Test button: it tells you which services you reach
"Out of space"	✨ menu → Settings → delete the old conversations
A section stays empty	Check the network: some sections load data from the internet
The charts are blurred	Reload the page; the charts are redrawn at your screen's resolution

Contacts

Author: Samuele Pio Provenzano **Repository and reports:** github.com/samupropio1-ship-it/BioSpecInfo-v11

Guide updated to version bsi-v181.

Test Documentation — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	bsi-v181
Purpose	Describe how the tests are organised, how to run them, what they cover and where they leave gaps.

1. Kinds of test present

Kind	Present	Notes
End-to-end (E2E)	✓	Playwright on headless Chromium against a local HTTP server. It is the prevailing form.
Data verification	✓	Scientific data compared against an independent source
Stability tests	✓	Long sessions, exhausted storage, degraded network
Documentation/code consistency	✓	Verifies that what the documentation promises really exists
Isolated unit tests	✗	See §6
Instrumented code coverage	✓	<code>audit_copertura</code> , with Chromium's profiler: no build, no source rewritten. See §6

Why E2E and not unit tests

The application is made of classic scripts operating directly on the DOM, with no modules and no dependency injection: there are no units that could be isolated without a deep rewrite. The defects that actually occurred — a blurred canvas, a button unreachable because its container is collapsed, an AI provider that stops answering — **would not have surfaced** from a unit test anyway: they live in the integration between code, DOM and browser.

The choice is therefore reasoned, not endured. Its limits remain, and are declared in §6.

2. How to run the tests

Requirements

```
npm install          # playwright-core only
python3 -m http.server 8899 &  # RDKit WASM requires an HTTP context
```

Without a server every test fails: opening the files over `file://` prevents the WebAssembly modules from loading.

The whole battery, with a report

```
node tools/genera-evidenza.js
```

It runs every bench, transcribes its full output and produces `docs/evidence/RAPPORTO-VERIFICA.md` with environment, commit and SHA-256 digests. It exits with code `0` only if no bench fails.

```
node tools/genera-evidenza.js --veloce # skips the benches that need a browser
```

The `--veloce` option is useful for a quick check but **is not sufficient for an attestation**: the benches it skips are precisely the ones measuring real behaviour.

A single bench

```
node tools/verifica-farmaci.js          # a stand-alone tool
node tools/verifica-farmaci.js --json   # machine-readable output

cd tools/banchi && node test_spettri.js # one E2E bench
```

The E2E benches live in `tools/banchi/`, versioned like the rest of the repository; the `BSI_BANCHI` variable allows another folder to be pointed at.

It has not always been so, and that is worth saying. Up to version `bsi-v168` the default folder was the working area of the session in which the benches had been written: **32 of the 39 benches were not in the repository**. Whoever cloned it and ran the documented command found seven, and received no warning — because a missing bench was counted but did not make the battery fail, which closed with "CONFORMANT" and zero failures.

Both things have been corrected: the benches are versioned, and **a missing bench makes the outcome NON-CONFORMANT**. A bench that is not there is not a bench that passed.

3. Composition of the battery

48 benches, grouped by what they demonstrate.

3.1 Scientific data

Bench	Verifies	Checks
verifica-farmaci	Structure ↔ molecular weight, duplicates, recorded deviations	233 entries
test_spettri	Functional-group recognition on reference molecules	36
test_spettri_ui	Grafting of the engine into the app's spectrum-drawing points	16
test_assi	Axis conventions in the SVG spectra	13
test_assi_canvas	Axis conventions in the canvas spectra	6
test_costanti	Physical-constant lookup and refusal of ambiguities	45
audit_dati	Consistency of the tabulated data	29
test_farm_ui	That the Drug Atlas draws every drug it holds in memory, category by category, and that a never-labelled category still appears	11
test_astro	The astrochemistry data: molecular weight ↔ formula for the 30 interstellar molecules against IUPAC atomic weights, no repeated celestial body, every quiz question with a valid answer among its options	13
test_datasci	The maths the Data Science section displays : R ² on an exact relation, standardised weights in the true ratio, ROC on hand-computable values, behaviour on pure noise, and that no section is left unreachable	23
test_simmetria	Point groups, normal modes, IR/Raman selection rules	26
test_cheminfo	The cheminformatics engine: Tanimoto and Dice against hand-computable values, Morgan and MACCS fingerprints, Butina clustering, Bemis–Murcko scaffolds, PCA, kernel ridge, scaffold split, null model by label scrambling , grouped cross-validation, CSV and SDF exports, contrast of the nine panels , agreement between lab and engine, fragmentation, matched pairs against known effects , substructure search, SAR table, model comparison in both directions, enrichment (EF and BEDROC) on hand-computable cases, conformal intervals and their refusal on classification	189

A bench that verifies a model must also verify it when the model is supposed to fail. `test_cheminfo` exercises the null model in both directions: on a learnable signal the true R² (0.394) beats all eight label-scrambled twins, with a margin of 0.287; on the **same molecules with random labels** the R² (−0.199) does **not** beat them. A check that verifies only the first case would pass even if the guard were wired to "true".

A "contrast 0" that did not cover the new surface. `verifica-accessibilita` measures what is visible, and the six panels of the Cheminformatics section stay hidden until the analysis has been run: it had never seen them. `test_cheminfo` now runs the analysis, opens the panels in turn and measures contrast over **460 text elements**, with the WCAG formulas rewritten inside the bench — asking the code under test how good its own contrast is, is not measuring. The bench also demands that more than 250 were measured: if the panels failed to open, the zero would be empty and the check would fail.

3.2 AI agent

Bench	Condition simulated
test_404	The provider has retired the model
test_503	Temporary overload
test_ko · browser_ko	Provider unreachable from the browser
test_firma	Opaque reasoning signatures to be returned
test_tetto	Token ceiling learned from the error
test_attesa · test_attesalunga	Timeouts and imposed waits
test_doppioinvio	Two overlapping submissions
browser_prova	Reachability measurement without a key
test_nucleo · browser_prov	Agentic loop and provider selection
caccia_ai	Malformed answers, truncated stream, 502 with HTML, Stop during the answer, panel closed halfway, hostile text

3.3 Stability

Bench	What it puts to the test
audit_stabilita	89 sections opened 5 times, full storage, corrupted data, bursts of clicks
audit_promesse	Rejected and unhandled promises, with a healthy network and a dead one
audit_quota	localStorage.setItem forced to fail on 10 pages
test_sw	Offline, degraded network, update while working
test_filemanager	Store unavailable, space exhausted, failed writes
test_visore3d	WebGL context consumption while browsing molecules

3.4 Interface

browser_reset · browser_proxy · browser_proxyui · browser_rdkit · browser_lab · browser_frontiera · test_aggiorna · test_guidaproxy · test_fluidita

test_fluidita — 15 checks. It measures how long the page stays *blocked* at every section switch, across all 89. It found three defects no other bench could see: the "3D PRO Viewer" froze the page for **797 ms** on the first click — the profiler attributes them to 3Dmol's first `render()`, which compiles the WebGL context's shaders; the **296 figures** in "Synthesis" (5,351 SVG nodes) were all drawn before anything was shown, while three are visible on opening; and **296 timers 40 ms apart** were scheduled when that section opened — eleven and a half seconds of wake-ups, each calling `getTotalLength()` on every stroke of every figure, including those off screen.

After the fixes: median switch **14 ms**, worst **162 ms**, **no section above 100 ms** (there were two, the worst at 1,166 ms).

Half the checks are the *opposite proofs*, because a page that builds nothing is instantaneous: the WebGL canvas must appear anyway, all 296 figures must exist shortly after, no frame may stay empty, printing must not come out mute, and the search must filter even cards whose figure is not yet born. And the bench counts the sections it traversed: on ten instead of 89 the timings would be excellent and the measurement worthless.

3.5 Security and accessibility

Bench	Verifies
verifica-sicurezza	API keys in tracked files, clear-text passwords, secrets in <code>wrangler.toml</code> , telemetry, scripts from external domains
verifica-accessibilita	WCAG AA contrast, accessible names, field labels, alternative text, heading hierarchy, <code>lang</code> attribute — across 13 pages
audit_mobile	That at 390 px the page does not scroll horizontally, across all 89 sections
audit_rete	That no user data leaves the device : a canary value seeded into 71 stores, the app used across 6 pages and 89 sections, the URL, headers and body of every request inspected
audit_copertura	How many bytes of JavaScript are actually executed while walking the application: 49.89 %, recorded and defended

The obstacle belonged to the tool, not to the problem. For three versions code coverage was declared unmeasurable, with this reason: "instrumenting would require introducing a build, which the application does not have". That was true of `c8` and `istanbul`, which rewrite the source before running it. But **Chromium collects it by itself**, inside the engine, without touching a byte of the file served:

`Profiler.startPreciseCoverage`, which Playwright exposes as `page.coverage.startJSCoverage()`. No build was needed: what was needed was noticing that the constraint belonged to the tools chosen.

The first result was 100 % on every file, including 3.6 MB of `index.html`. That was the bench measuring nothing, in its most insidious form: a perfect number. V8 emits, for each function, an outer range with the number of entries, and inside it ranges with `count: 0` for what was not executed. Summing the positive ranges sums the whole file. One does the opposite: start from the total and subtract the union of the empty ranges.

The real number is **49.89 %**, and it holds the same pact as the accessibility debt: recorded, and the bench fails if it falls. An absolute threshold ("80 %") would be an invented figure; "it must not get worse" is a promise that can be kept.

Why a canary and not a list of requests. SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit: empty telemetry variable, no script from an external domain. That is real evidence but indirect: it says "I cannot see how it would leave", which is not "it did not leave". Looking instead at the list of hosts contacted would be just as weak, because it concludes from absence of evidence.

The canary reverses the burden. An unrepeatable string is written into the user's own data — notes, chats, API keys, settings, progress — then the application is **used**, and every outgoing request is opened and read. If the string appears nowhere, it is because it really did not leave.

```
BSI_PROVA_FUGA=1 node tools/banchi/audit_rete.js # must FAIL
```

The bench carries its own proof of working: with that variable it causes a leak on purpose, towards a **declared** host — contacting it is legitimate, sending it the user's data is not. If it does not fail then, the instrument is the thing that is broken.

Why a bench on horizontal overflow. UI-03 says "the app must work at 390 px", and the matrix gave it as verified by `audit_stabilita`, which in a phone viewport looks at **JavaScript errors**: a page can have not a single one and still run a hundred and sixty-three pixels off the screen. It was a requirement declared covered and not measured. The bench measures the document's `scrollWidth` — exactly what makes the scrollbar appear — and not any element wider than the screen, which inside a container built to scroll is correct and would fill the output with noise until nobody reads it any more.

What the accessibility check does not cover, and it must be said: text inside SVGs (the colour comes from `fill`, the background is a drawn shape) and text on gradient backgrounds (they do not have a colour). The skipped elements are counted and reported, not hidden.

One section at a time. The inspection skips elements that are not visible — and rightly so: a hidden element has no contrast to measure. But `index.html` alternates 89 sections and shows only one: out of **19,751** text elements the bench was looking at **41**, and printing "0 defects". It was not a false result, it was a result on a sample nobody had declared. The sections are now opened one by one, and **the number of sections traversed is printed**.

That count earned its keep on the very first run: the first attempt traversed **zero** — Playwright's click waits for the element to be visible, and the buttons sit inside collapsed navigation groups — and the bench said so ("87 sections present, none traversed") instead of printing another reassuring zero.

3.5-bis The accessibility debt, and the pact that it must not grow

Walking every section, **1,069** contrast defects surfaced. Today they are **zero** — the list of causes and remedies is in docs/09 §4 — and unlabelled fields have fallen from 40 to **zero**. The mechanism described below remains, and matters more than the number it guards. Demanding zero *from the outset* would have left only two roads, both bad: a verification red for ever, or one loosened until it turned green again. The third road is to **declare the number** and forbid it from growing, whatever it is. That is how 1,069 became zero: every step recorded, none of them reversible in silence.

File	Role
docs/evidence/accessibilita-riferimento.json	The recorded count

Situation	Bench outcome
The count rises	✗ FAILED — it is a regression
The count falls	✓ passes, and asks for the baseline to be updated
The count is unchanged	✓ passes silently: debt declared, not grown — today that value is 0 , and the bench defends it

```
node tools/verifica-accessibilita.js --aggiorna-riferimento
```

To be used **only after reducing** the defects, never to turn a regression green. Even at zero this is not full WCAG conformance: text inside SVGs and text over gradients stay outside, which automation cannot judge and which the bench **counts and reports** on every run (docs/09 D-09). It is the honest measure of what can be measured, with the guarantee that it will not silently get worse.

3.6 Consistency

Bench	Verifies
verifica_guida	That the documentation's promises exist in the code; absence of conflict markers across 58 files
verifica-documenti	Internal links, version alignment, index consistency, justification of the deviations, internal consistency of the traceability matrix (no duplicated identifiers, coverage totals equal to the identifiers actually defined)
genera-pacchetti	Rewrites the references of the extracted documents and verifies that no link is left broken in the three delivery packages
genera-pdf	Converts the documents to PDF and fails if a file falls below the plausibility threshold (empty conversion)
verifica-affermazioni	Compares every number declared in the documents — sections, drugs, diseases, tumours, strategies, modules — with the one measured in the running application , in Italian <i>and</i> in English

Why numbers must be measured, not remembered. The technical dossier opens with figures: "63 diseases (23 tumours)", "46 strategies", "25 modules". They are the first thing an assessor reads, and they are enough for everything else to be believed or doubted. Nobody was checking them: written once, they stayed put while the application changed. And indeed `chimorga.html` declared **"25 modules"** at the top and **"17 modules"** further down — two statements about the same object, in the same file, one of them false.

A lesson paid for while writing the bench. I had already concluded that "63 diseases" was an exaggeration: in the source I had found a `DISEASES` array with 12 entries, and I was about to "correct" the document. Measuring in the DOM I saw that the menu really does list 63 — the array I had found was a different, smaller one. **A count on the source is not a measurement: it is a clue.** That is why the bench opens the page, opens the section, and counts.

It also checks that the documents agree **with one another**: two documents saying different things about the same object are a defect even when one of them is right. The English translation counts as one of those documents — `docs/en/05` had been left at "84 sections" while the app had 87, and had been stuck at version `bsi-v146` for three releases — and no bench saw it, because none looked at the English set.

Why a check on the matrix. The matrix declared `SCI-10` twice — once as a requirement verified at 100 %, once as an explicitly **unverified** one — and the summary table added up to 36 requirements while 37 were defined in the document. Both errors sat in the number an assessor reads first. The bench now counts the identifiers actually present and fails if the totals written do not match: the proof that the check really measures something is that, with the two errors reintroduced one at a time, it reports both.

4. Notable test cases

Some benches deserve a mention because they verify properties a functional test does not observe.

Case	Method	Measured result
Memory leaks	89 sections × 5 rounds, DOM nodes counted each round	+26,876 on the first round (construction), +0 on the four that follow
Exhausted storage	<code>setItem</code> replaced with a function that always throws	10 pages out of 10 stay operational
Degraded ≠ absent network	Requests held for 20 s, interception at context level	Answer from cache in 3,507 ms (threshold 3,500)
Spectrum reproducibility	Drawn twice, compared byte for byte	Identical
WebGL contexts	Viewer constructions over 10 consecutive molecules	From 7 to 1

Case	Method	Measured result
Corrupted history	Two Enter presses 500 ms apart	user,assistant instead of user,user,assistant,assistant
Text contrast	WCAG formula on every element with text of its own, 13 pages and 89 sections	19,751 elements examined (it was 41): 1,069 defects surfaced, 1,069 corrected , 0 recorded as a value that must not grow
Immediate cancellation	Stop pressed at 1.5 s, state sampled every second	Send button available from the 1st second (it was the 10th)

5. How to add a bench

Minimal structure

```

/* A comment explaining WHAT this bench demonstrates and why it is needed. */
const { chromium } = require('playwright-core');
const CHROME = '/opt/pw-browsers/chromium-1194/chrome-linux/chrome';

(async () => {
  const b = await chromium.launch({ executablePath: CHROME, args: ['--no-sandbox'] });
  // serviceWorkers:'block' if the bench modifies files: without it the SW
  // might serve a cached copy and the old file would be measured
  const ctx = await b.newContext({ serviceWorkers: 'block' });
  const pg = await ctx.newPage();
  const err = [];
  pg.on('pageerror', e => err.push(e.message));

  let ok = 0, ko = 0;
  const att = (d, atteso, avuto) => {
    if (String(atteso) === String(avuto)) { ok++; console.log(' ✓ ' + d + ' → ' + avuto); }
    else { ko++; console.log(' x ' + d + '\n      expected: ' + atteso + '\n      got:      '
+ avuto); }
  };

  await pg.goto('http://127.0.0.1:8899/index.html', { waitUntil: 'load', timeout: 60000 });
  await pg.waitForTimeout(2500);

  // ... measurements ...

  att('no JS error', 0, err.length);
  await b.close();
  console.log('\n' + (ko ? 'x ' + ko + ' FAILED, ' : '') + ok + ' passed');
  process.exit(ko ? 1 : 0); // the exit code is what counts
})();

```

The name is then added to the appropriate family in `tools/genera-evidenza.js`.

The rule that matters

A bench that measures nothing passes. It is the most insidious way to obtain a false guarantee: no alarm, no failure, and no verification.

Therefore **every bench that simulates a condition must count how many times the simulation actually fired, and fail if that count is zero.**

Traps actually met while building the battery:

Trap	How it showed itself
<code>page.route</code> does not intercept Service Worker requests (<code>context.route</code> is needed)	The "degraded network" test measured a healthy network and concluded in 16 ms instead of 3,500
Probing exhausted <code>localStorage</code> with a one-byte write	After saturation one byte always finds room: the bench "passed"
Looking for <code>[data-tab]</code> where navigation uses <code>[data-s]</code>	Zero tabs found, therefore zero measurements — and zero failures
The ticks of an axis on <code><canvas></code> are not text	They must be collected by intercepting <code>fillText</code> ; and the bottom-most line is the axis <i>title</i> , not a tick
A probe installed on a library loaded afterwards	It is replaced along with the library: zero calls counted while the code was working
A <code>var</code> inside an IIFE is not <code>window.something</code>	Assigning it from outside creates a different variable

6. Coverage gaps

Declared. The coverage reported in `08-Traceability-Matrix.md` measures **how many requirements have a bench**, not how many lines of code are executed: they are two different things and must not be confused.

Gap	Current situation	Recommendation
Code coverage	Measured: 49.89 % of statements over the widest path. What stays outside is whole-battery coverage and branch coverage: an <code>if</code> entered from one side only counts as covered	Extend collection to every bench, and move from statement coverage to branch coverage
Accessibility	Automated over 13 pages and all 89 sections: WCAG contrast, accessible names, labels, alternative text, heading hierarchy. Text inside SVGs and over gradients stay outside, and are counted on every run	Add <code>axe-core</code> alongside, for the rules this bench does not implement (ARIA roles, tab order, focus management)
Security	<code>verifica-sicurezza</code> runs 9 checks over 235 tracked files and covers SEC-01, SEC-03, SEC-05, SEC-06; SEC-04 is covered by <code>verifica_guida</code> . SEC-02 remains indirect : see <code>docs/09 D-03</code>	Observe the network traffic during real use, the only direct verification of SEC-02
Browsers other than Chromium	No automatic test on Firefox or WebKit	Extend the main benches to <code>webkit</code> , where the differences on IndexedDB and Service Worker are greatest
Performance	Manual cross-device testing	Automatic measurement of first-paint time

Gap	Current situation	Recommendation
Visual regression	Absent	Screenshot comparison for the charts, which are the heart of the product

7. Acceptance criteria

A version is not published if even one of these is unsatisfied.

- ❑ `node tools/genera-evidenza.js` closes with **0 failures**
- ❑ `tools/verifica-farmaci.js`: no defect, every deviation recorded with its reason
- ❑ `verifica_guida`: no promise in the documentation without a counterpart
- ❑ No non-network JavaScript error on any page
- ❑ Evidence report regenerated against the commit being published

Document updated to version `bsi-v181`.